1. Introduction and Motivation

1.1. What's in a Name?

1.2 .Wireless Communication

1.3. The Internet – Overview and Philosophy

1.4. Hierarchy Principle and Layer Models

1.5. Packet Sniffing with WireShark

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1.1. What's in a Name?



Kommunikation (lateinisch *communicatio* ‚Mitteilung') ist der Austausch oder die Übertragung von Informationen, die auf verschiedene Arten (verbal, nonverbal und paraverbal) und auf verschiedenen Wegen (Sprechen, Schreiben) stattfinden kann, auch im Wege der computervermittelten Kommunikation.

„Information" ist in diesem Zusammenhang eine zusammenfassende Bezeichnung für Wissen, Erkenntnis, Erfahrung oder Empathie. Mit „Austausch" ist ein gegenseitiges Geben und Nehmen gemeint; „Übertragung" ist die Beschreibung dafür, dass dabei Distanzen überwunden werden können, oder es ist eine Vorstellung gemeint, dass Gedanken, Vorstellungen, Meinungen und anderes ein Individuum „verlassen" und in ein anderes „hineingelangen". Dies ist eine bestimmte Sichtweise und metaphorische Beschreibung für den Alltag – bei genaueren Beschreibungen des Phänomens *Kommunikation* wird die Anwendung dieser Metapher allerdings zunehmend schwieriger.

(http://de.wikipedia.org/wiki/Kommunikation, 29.9.2024)

Ein **Verteiltes System** ist nach der Definition von Andrew Tanenbaum ein Zusammenschluss unabhängiger Computer, die sich für den Benutzer als ein einziges System präsentieren. Peter Löhr definiert es etwas grundlegender als „eine Menge interagierender Prozesse (oder Prozessoren), die über **keinen gemeinsamen Speicher** verfügen und daher **über Nachrichten miteinander kommunizieren**".

(http://de.wikipedia.org/wiki/Verteiltes_System, 29.9.2024)

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1.2. Wireless Communication









The “beacon of Amon Din” in the movie “Lord of the Rings – The Return of the King”.

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Optical Telegraphs

The age of telegraphs began in **1793**, when Frenchman Claude Chappe developed the first (optical) telegraph.

The so-called **"semaphore-telegraph"** has

- a **"regulator"** (large arm) and
- two **"indicators"** (short arms).

Greek:
sema: sign, see "semantics"
pherein: carry, see e.g. "amphora"

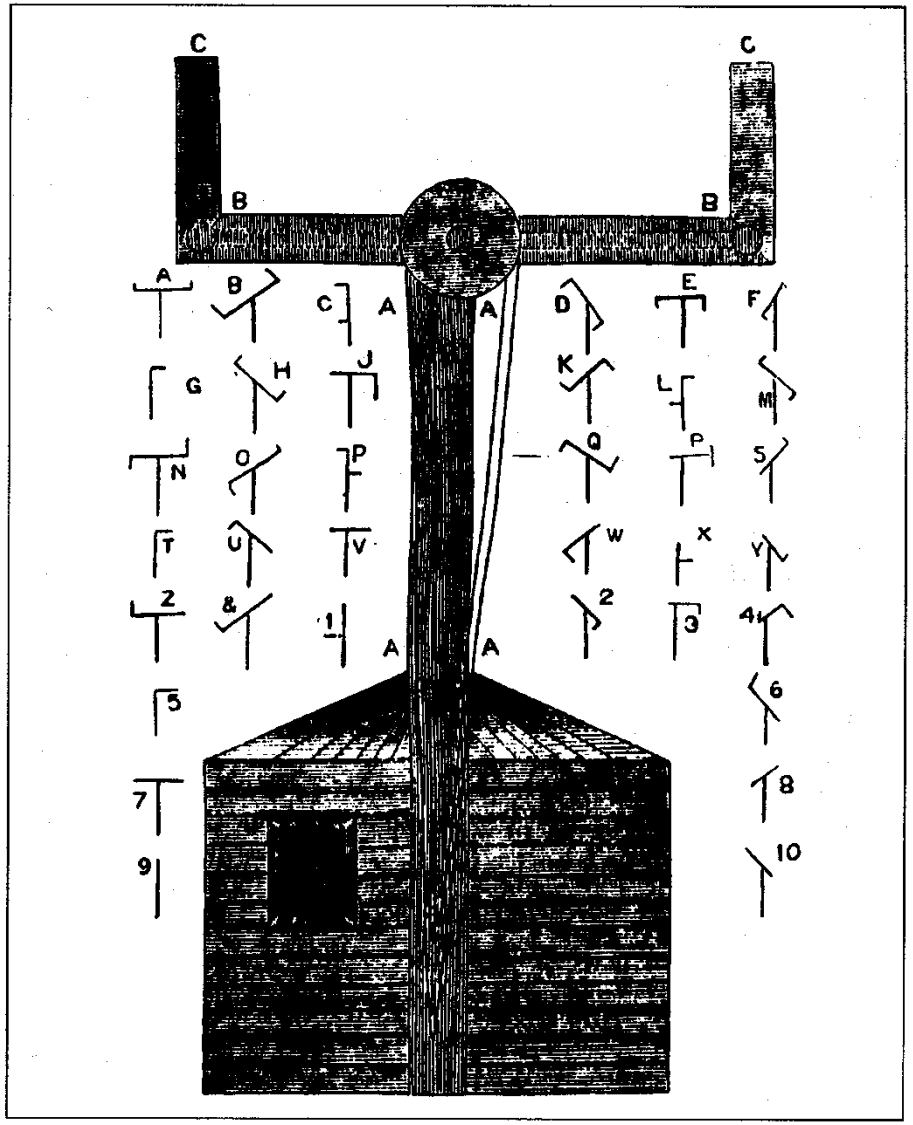


Figure A.1 The French Telegraph.
(Source: Gentleman's Magazine, Nov. 1794, p. 992)

Source:
G.J. Holzmann, B. Pehrson, "The Early History of Data Networks", IEEE 1995, ISBN 0-8186-6782-6

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France 1793: Construction of Optical Transmission Lines

12.7.1793 **First official test** (successful!)

26.7.1793 The Paris parliament decides to establish a **national telegraph authority**.

4.8.1793 The parliament provides **58,400 francs** for the construction of the optical telegraph between **Paris** and **Lille** (former northern frontier, approx. 190 km from Paris)

24.9.1793 The parliament approves the erection of telegraphs **on top of bell towers, other towers and other suitable facilities.** It is approves the **removal of trees** in the range of optical transmission lines.

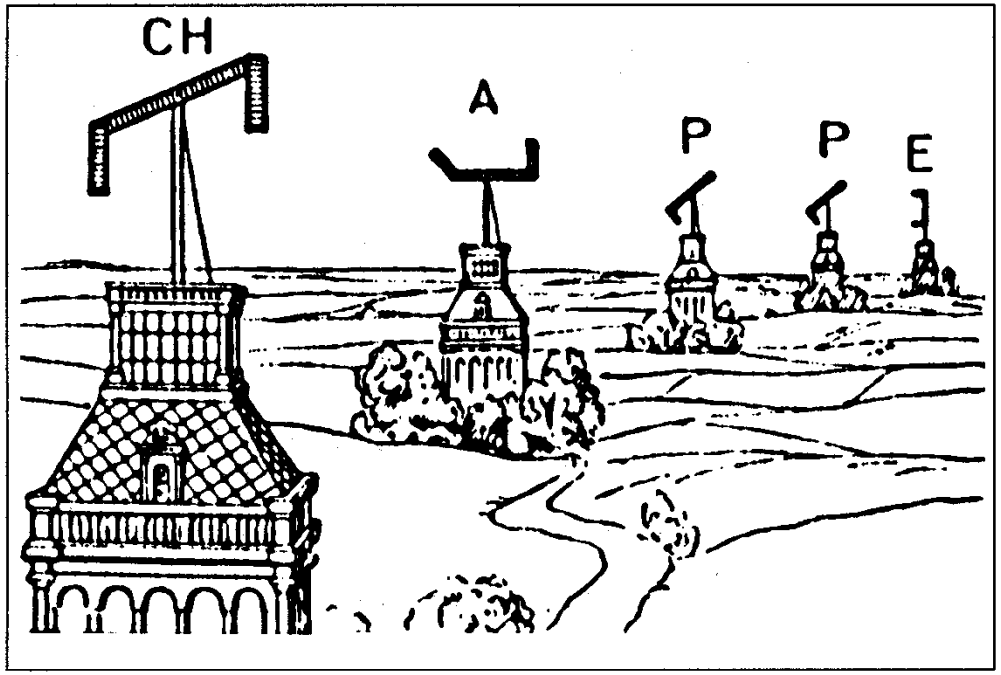


Figure 2.16 Transmission Line.
(Source: [Koenig 1944], p. 435)

Source:
G.J. Holzmann, B. Pehrson,
"The Early History of Data Networks", IEEE 1995, ISBN 0-8186-6782-6

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The Semaphore Telegraph Network in France (1794-1846)

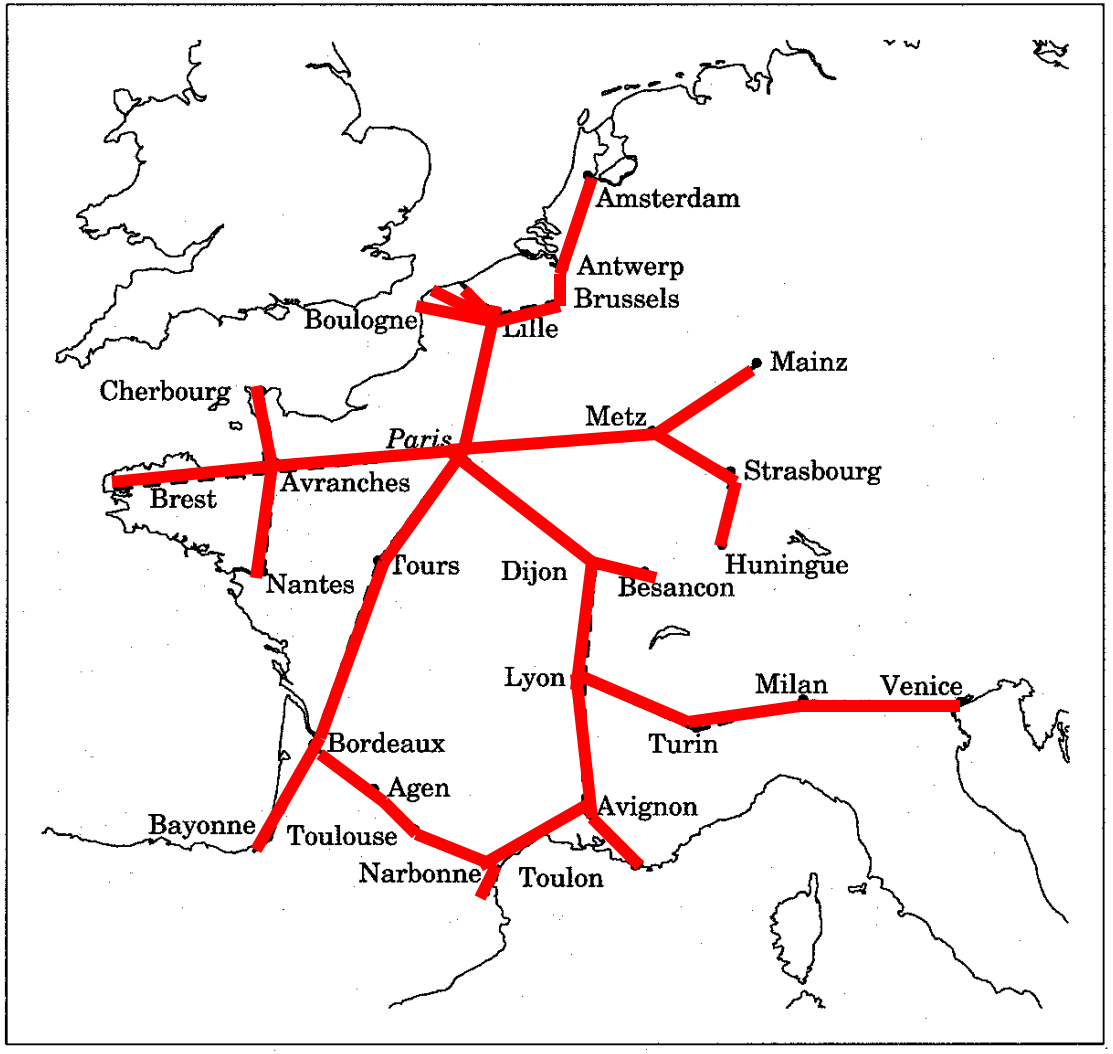
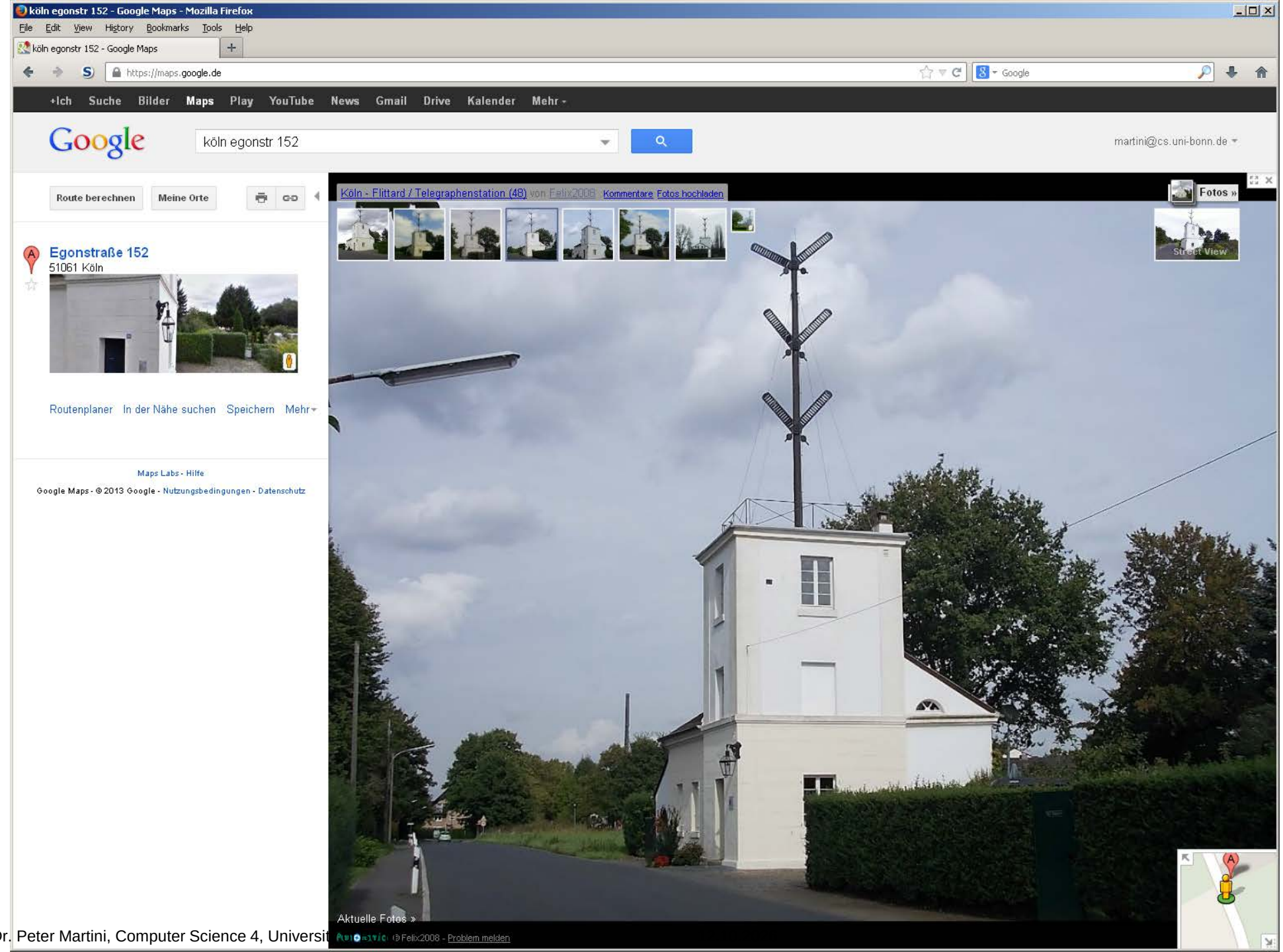


Figure 2.14 The Semaphore Telegraph Network
Overview of Lines Constructed between 1794 and 1846.

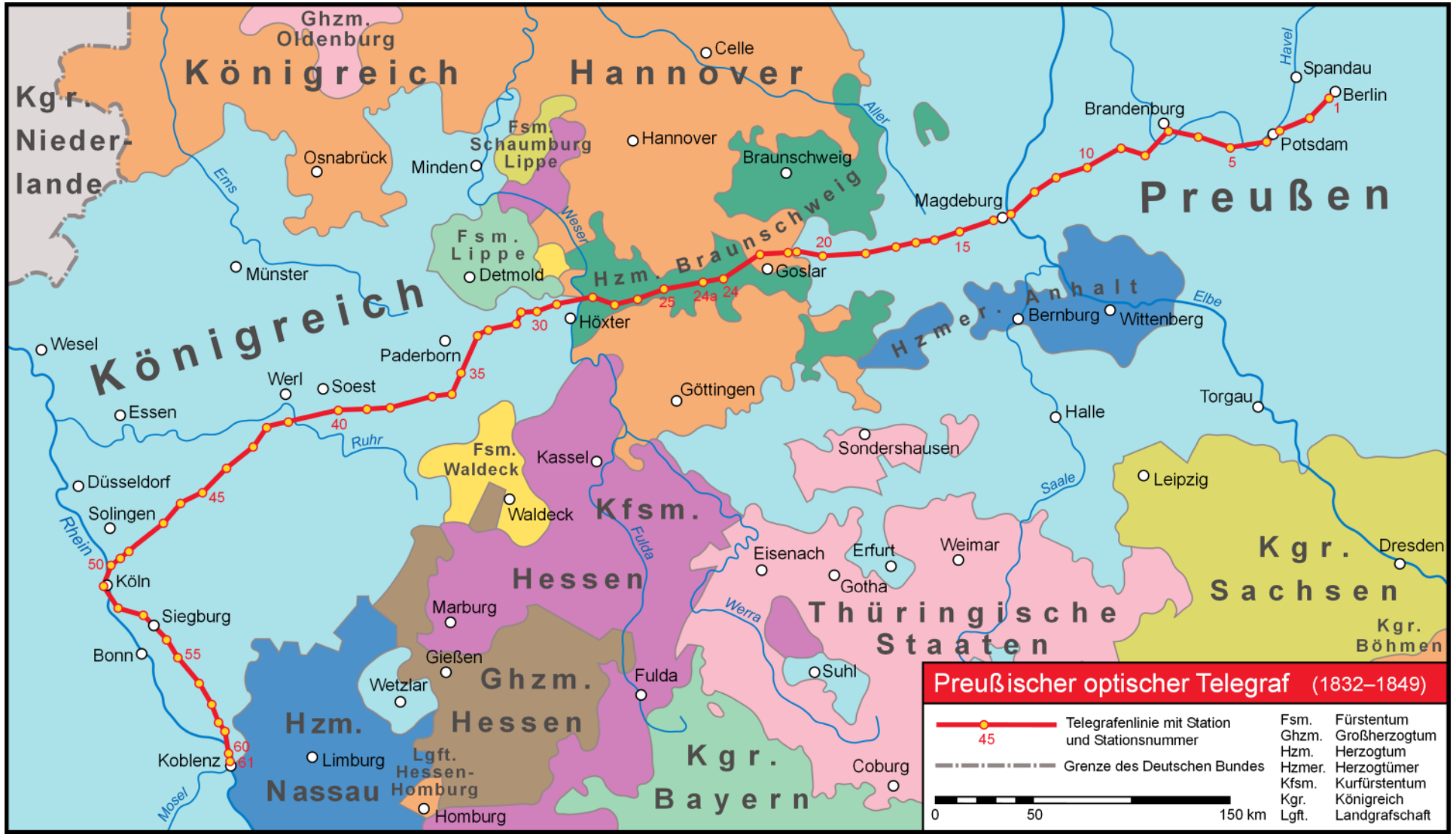
Source:
G.J. Holzmann, B. Pehrson,
"The Early History of Data Networks",
IEEE 1995, ISBN 0-8186-6782-6

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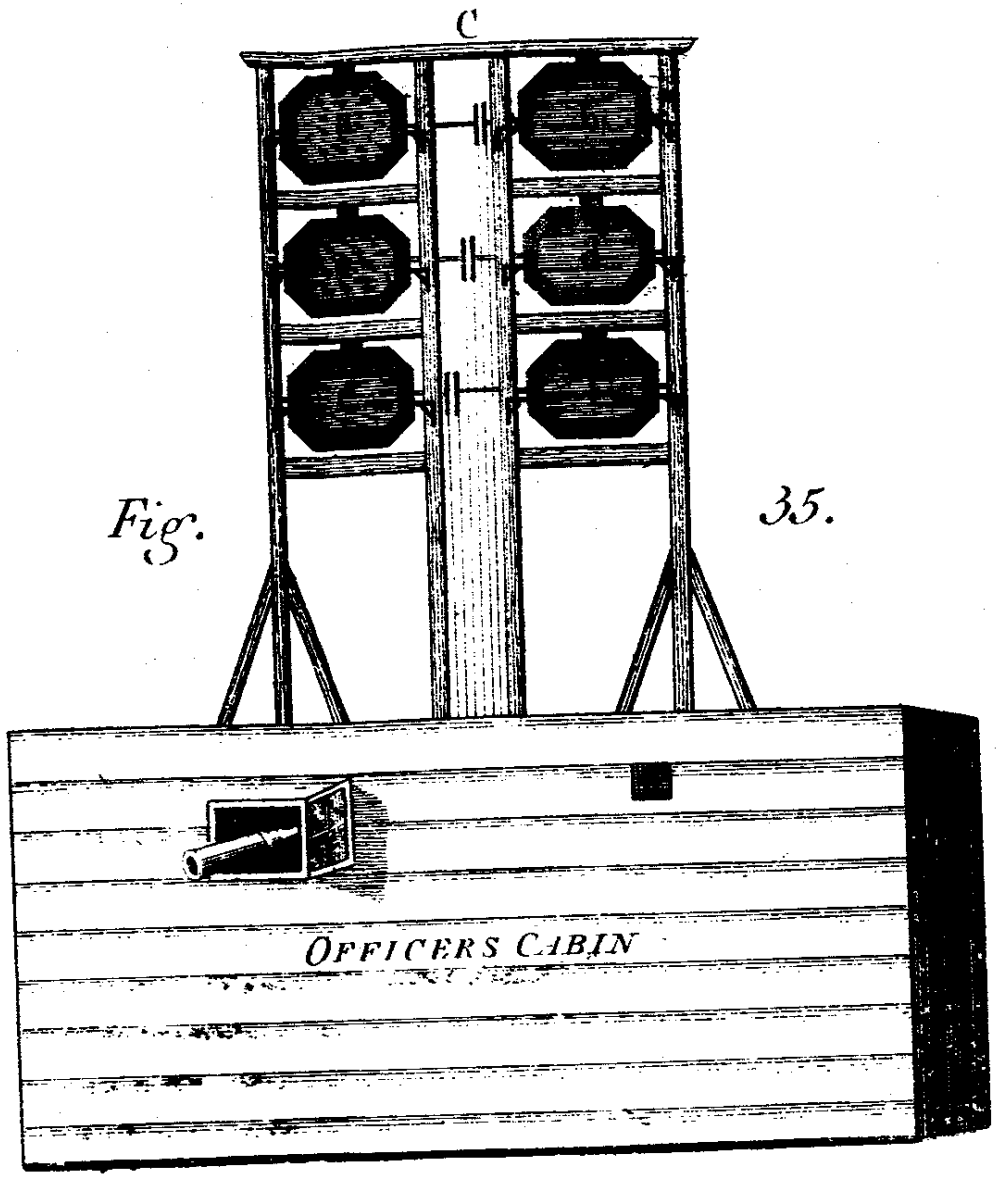
1832-1849: The Line from Berlin to Koblenz



Source: http://de.wikipedia.org/wiki/Preußischer_optischer_Telegraf

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Optical Telegraphs in England



Source:
G.J. Holzmann, B. Pehrson,
"The Early History of Data Networks",
IEEE 1995, ISBN 0-8186-6782-6

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Optical Telegraphs in England (2)



Figure 5.9 Working the Ropes, Murray's Design.
(Source: [Holmes 1983])

Source:
G.J. Holzmann, B. Pehrson,
"The Early History of Data Networks",
IEEE 1995, ISBN 0-8186-6782-6

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Optical Telegraphs in the Netherlands

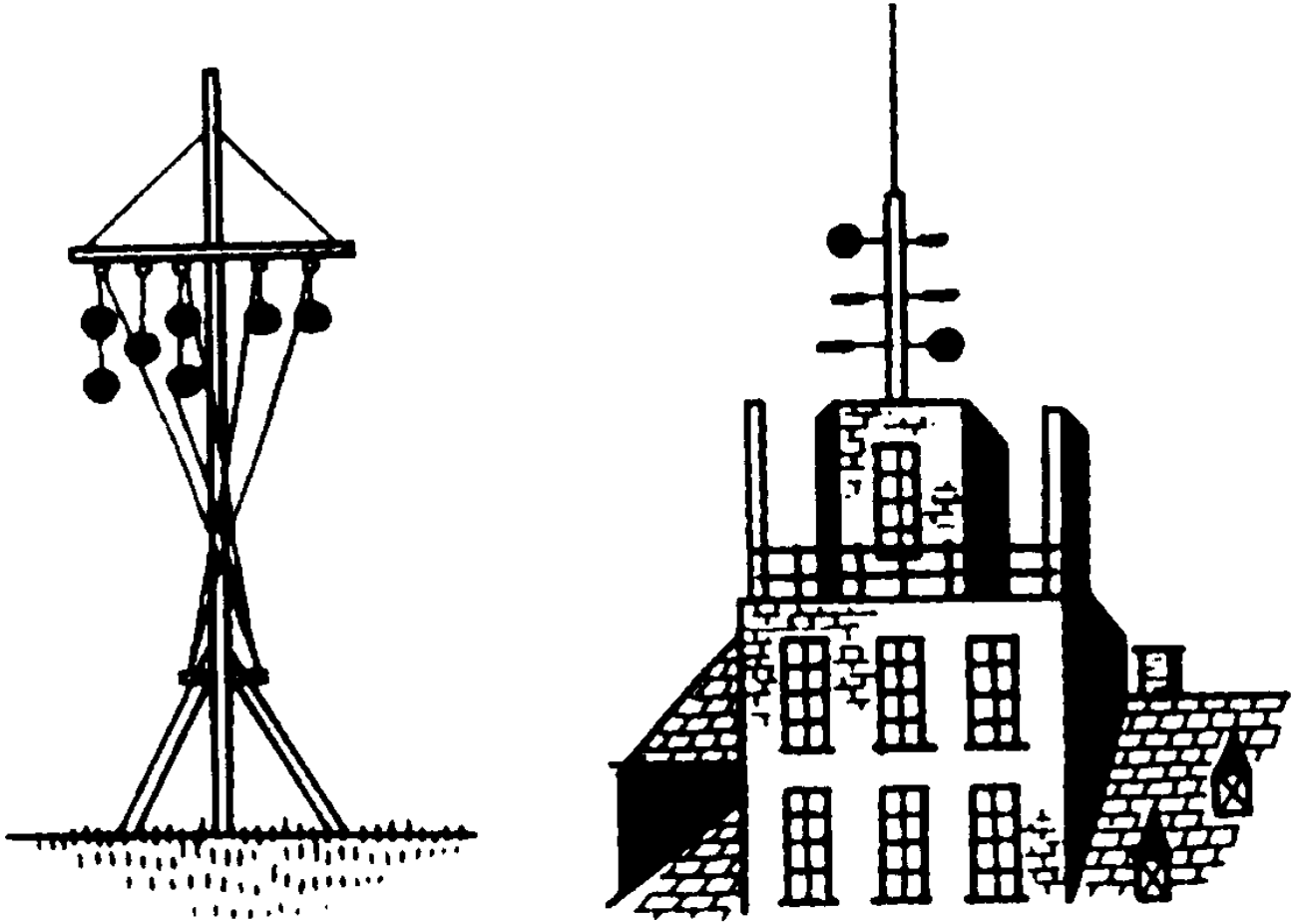


Figure 1.12 Dutch Semaphores—Nineteenth Century.
(Source: [Ringnalda 1902], p. 1)

Source:
G.J. Holzmann, B. Pehrson,
"The Early History of Data Networks",
IEEE 1995, ISBN 0-8186-6782-6

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1.3. The Internet – Overview and Philosophy

The majority of today's (technical) communication systems follows the "Internet Philosophy", … at least to a certain extent …

In this section, we provide an overview of Internet essentials.

1.3.1. How does the Internet work?

1.3.2. Packet Switching

1.3.3. The first "Internet"

1.3.4. The "Transmission Control Protocol" (TCP)

1.3.5. The "Internet Protocol" (IP)

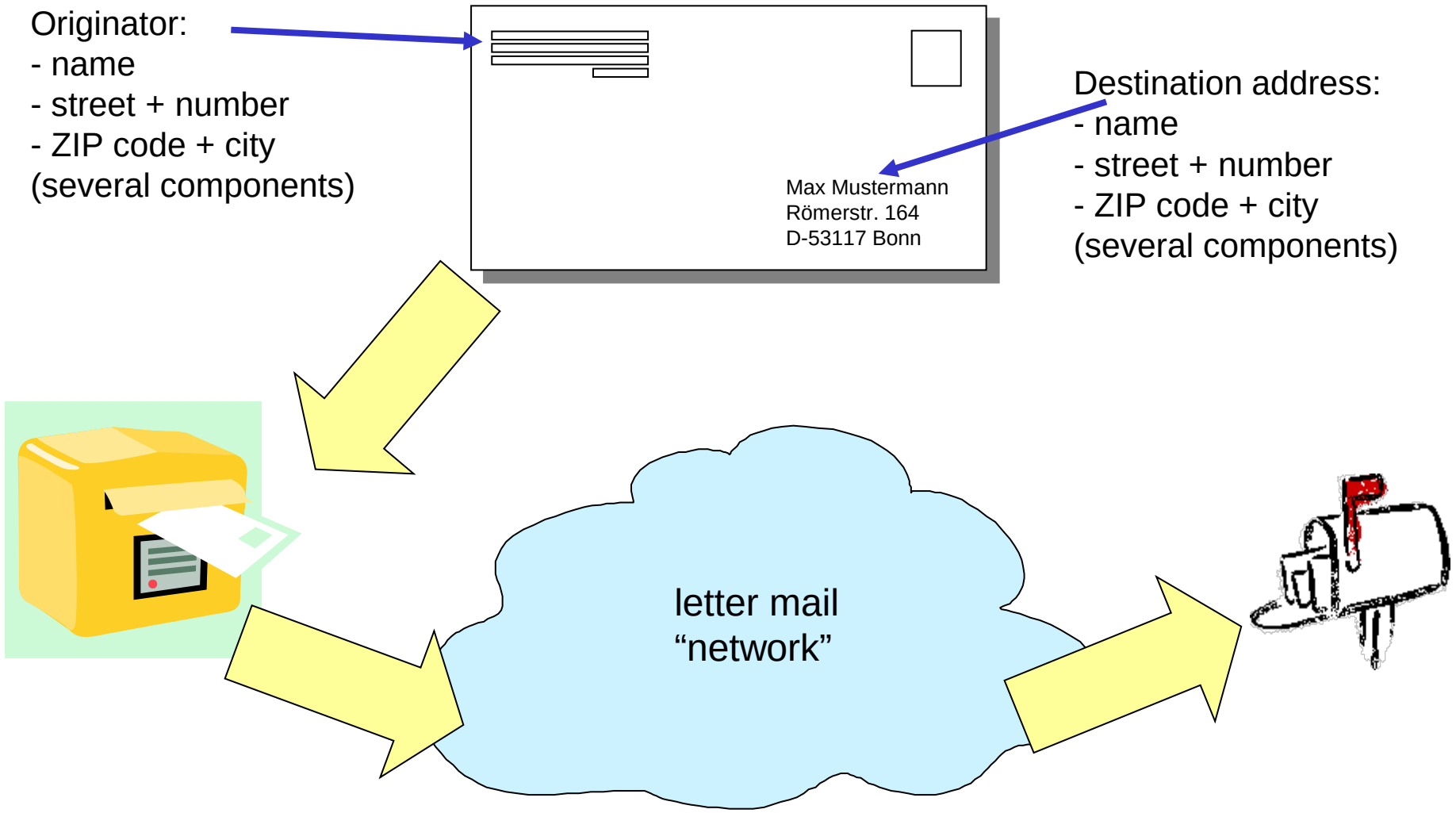
1.3.6. The TCP/IP-Internet

1.3.7. A Network of Networks

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1.3.1. How does the Internet work?

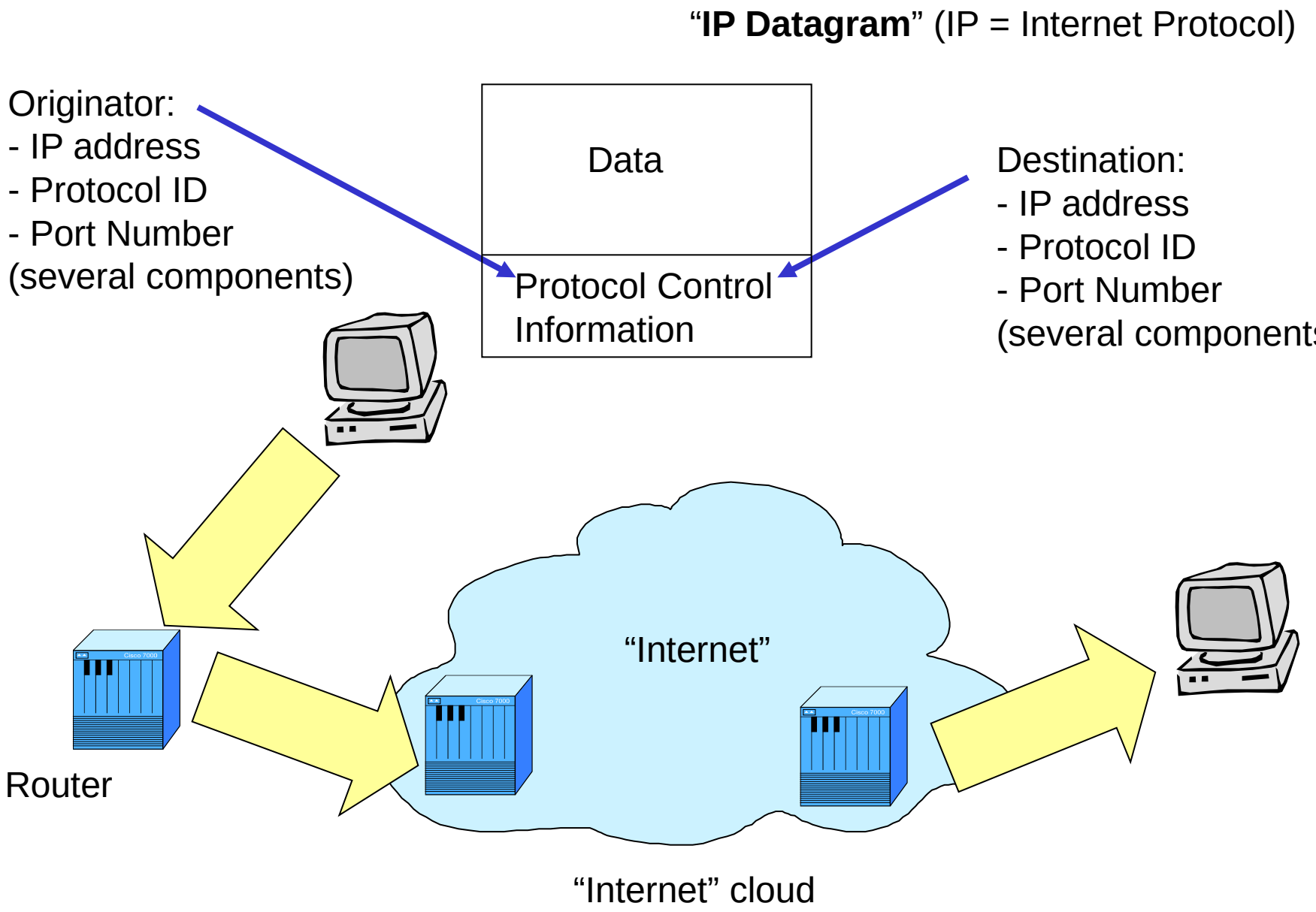
Analogy: letter mail



abstract “network” cloud

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How does the Internet work?

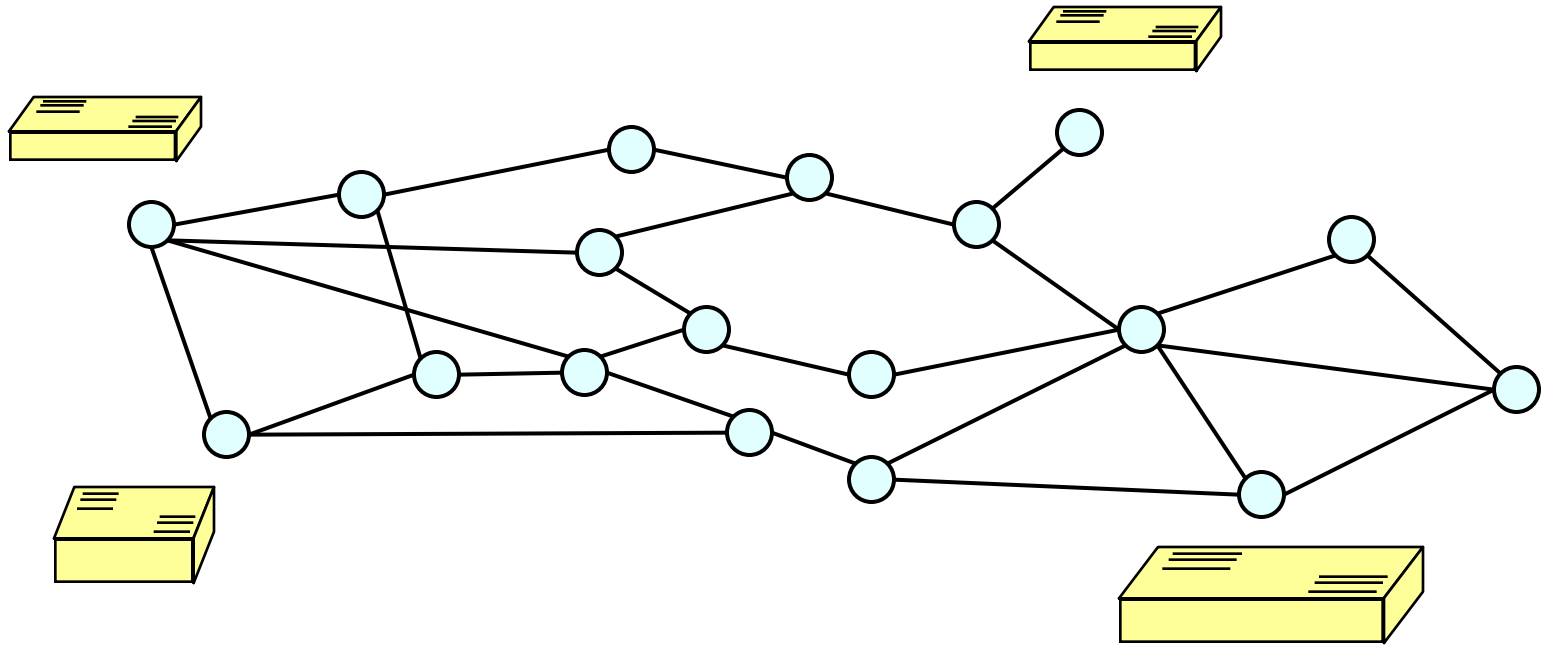


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1.3.2. Packet Switching

In the Internet, the information units (packets, datagrams) travel across a **store-and-forward network of redundant connections**:

Packets are stored until they have been forwarded to the next-hop station.

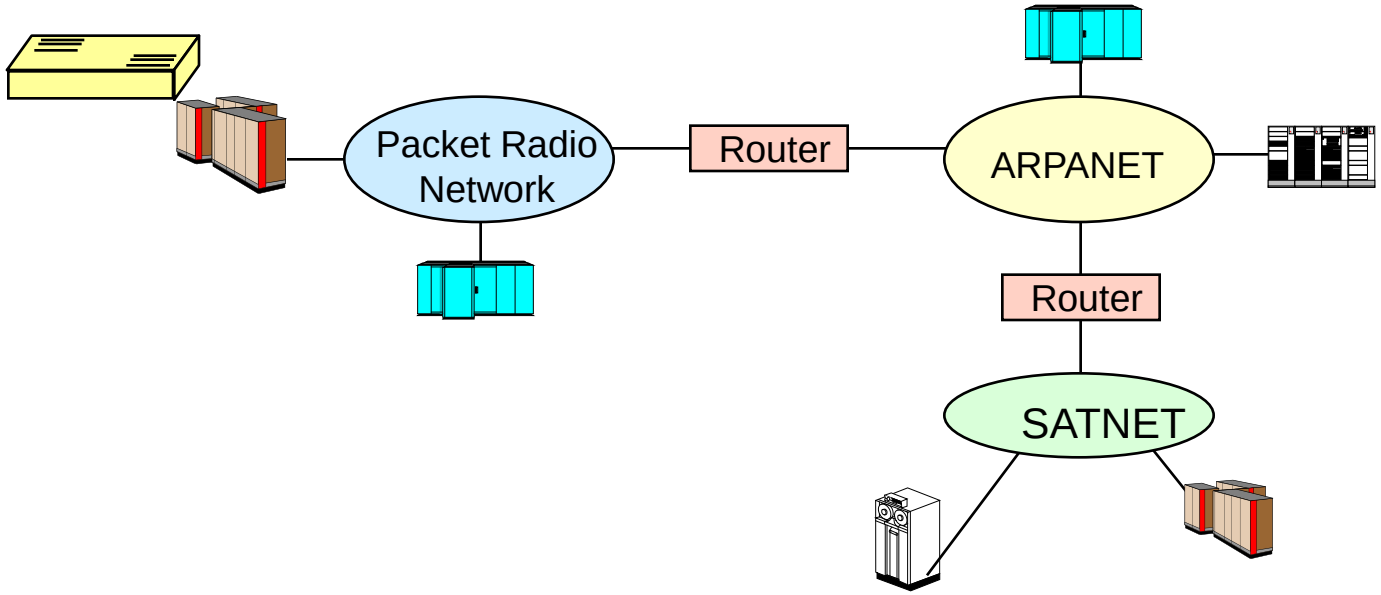


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1.3.3. The first “Internet”

Packet Switching may be used for forwarding packets across …

- ... a **specific network**
- … a **network of networks** (internetwork, “internet”).

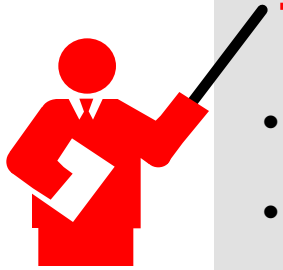


The first “internetwork” became reality in October 1977:

- Packet Radio Network in San Francisco
- ARPANET across the USA
- SATNET via satellite across the Atlantic to London.

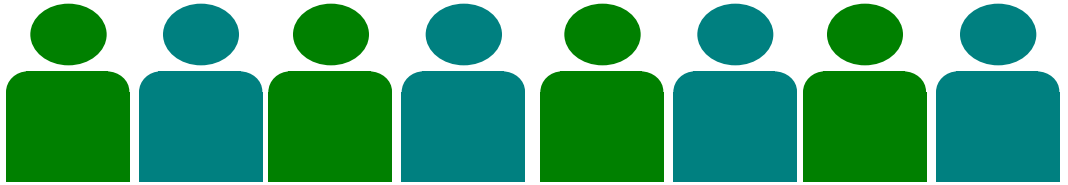
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1.3.4. The “Transmission Control Protocol” (TCP)



The Internet Philosophy

- Networks are unreliable.
- End systems take care of error detection / correction.



The **Transmission Control Protocol (TCP)**

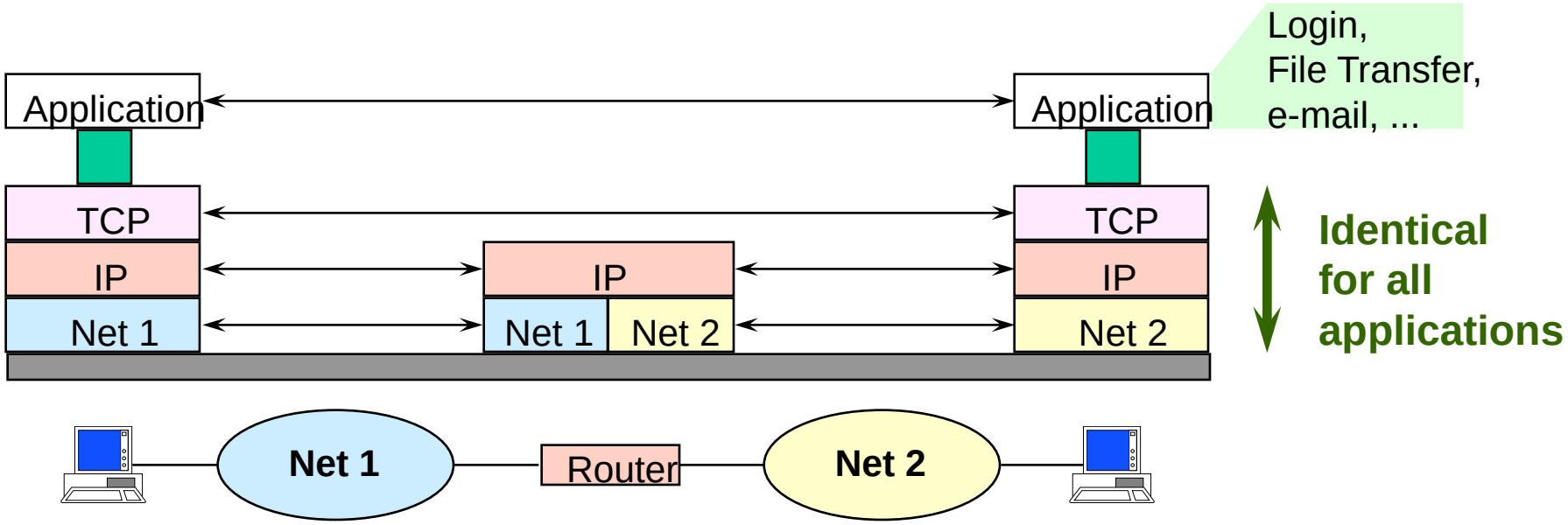
- **makes sure that all damaged and lost packets are retransmitted** and that **duplicates are removed**,
- **re-orders** the messages at the receiver (ordering preservation),
- **splits large information units** into flows of small, numbered packets,
- **decides how fast packets are sent to the network,** subject to capabilities of the sender, the receiver and the network in-between.

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1.3.5. The “Internet Protocol” (IP)

The **Internet Protocol (IP)**

- **specifies what all packets must look like** to allow for “routing”
  (based on hierarchic addresses: network-ID, subnetwork-ID, ..., host-ID).

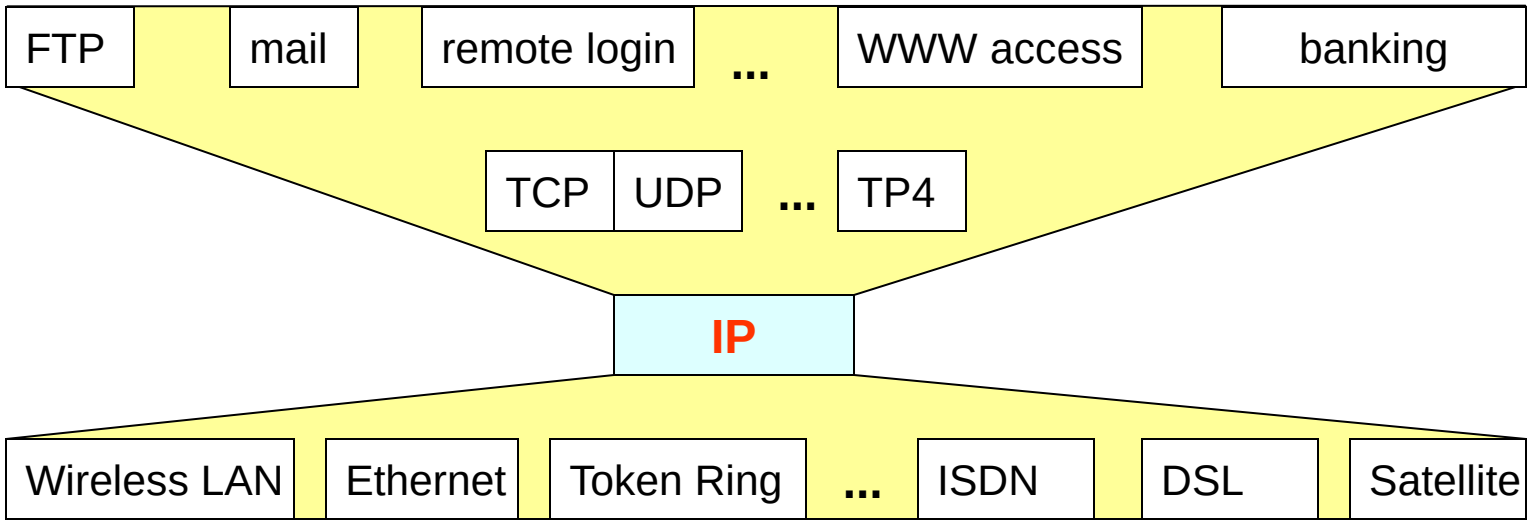


The **Internet Protocol**

- **defines** ..... un-ambiguous, standardized, network-independent addresses,
- **tries** ..... to take the messages to the receiver **somehow**,
- **does not guarantee anything.**

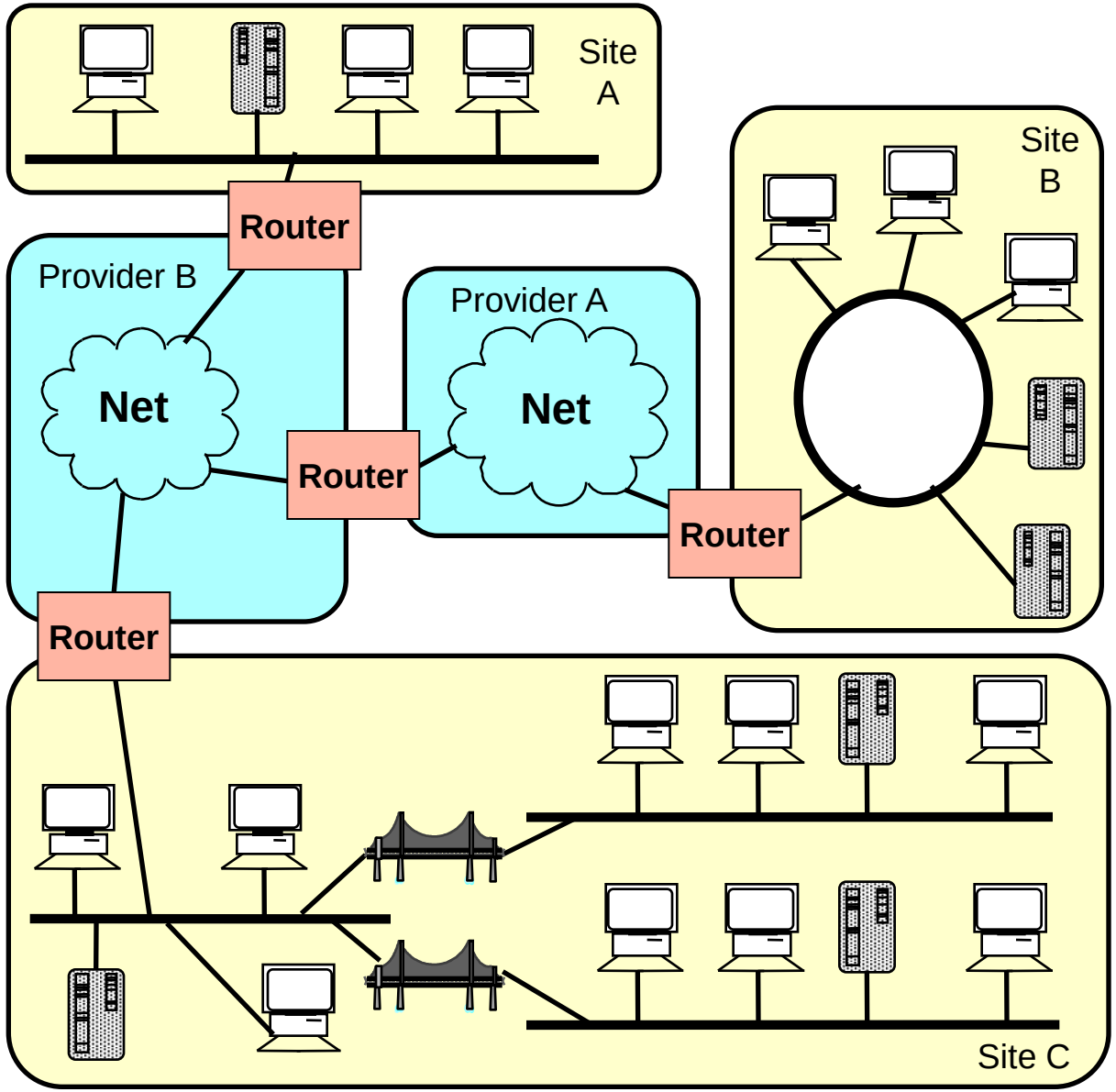
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IP: The Slim Waist



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1.3.7. A Network of Networks



a) central: “Provider”

Provider (different Organizations, regional or country-wide) interconnect their networks forming a world-wide backbone network.

b) peripheral: “Sites”

Sites (networks subject to a central administration) connect their networks to the providers.

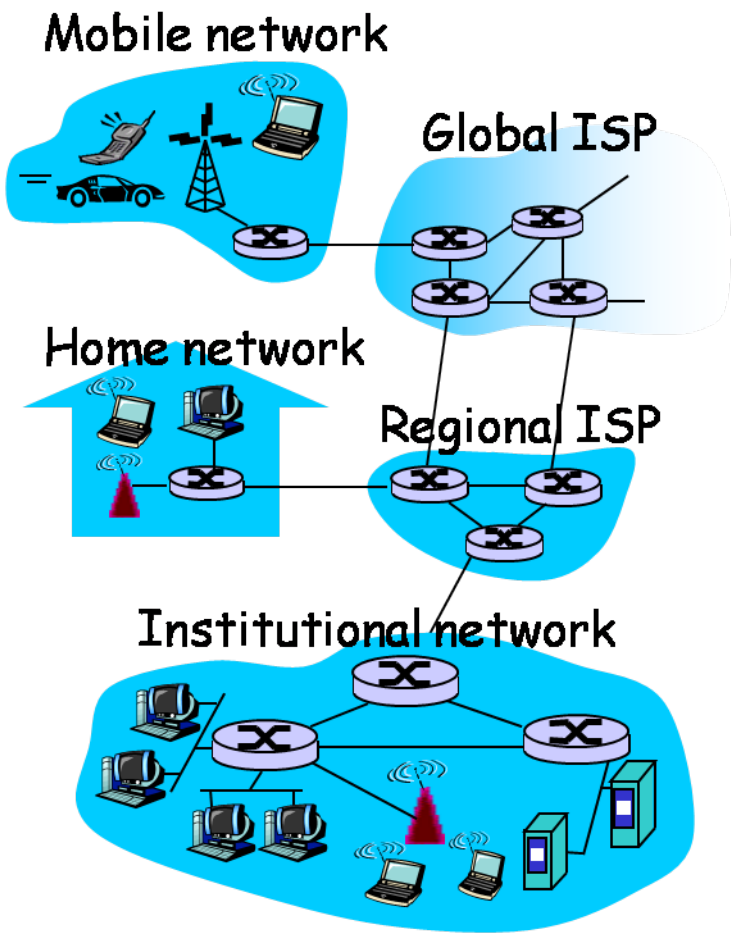
The simplest case: A site connected to a provider.

Examples for sites: Universities, companies.

Remark:
Sites are also called
“**autonomous systems**”

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A “nuts and bolts” view

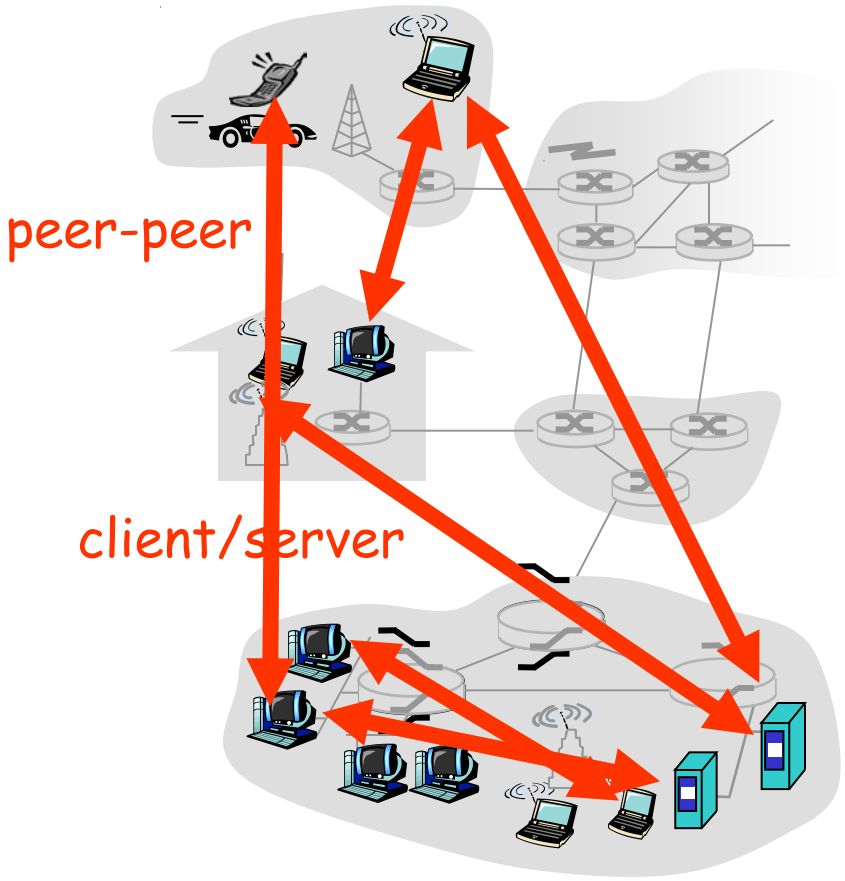


Source: Jim Kurose, Keith Ross: Computer Networking: A Top Down Approach - 5th edition, Addison-Wesley, April 2009.

- millions of connected computing devices: ***hosts*** *(end systems)*
  - running ***network apps***
    - Web, VoIP, email, games, e-commerce, file sharing
- ***communication links***
  - fiber, copper, radio, satellite
  - different transmission rates and properties
- **routers**
  - forward packets (chunks of data)

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Network Edges and Services

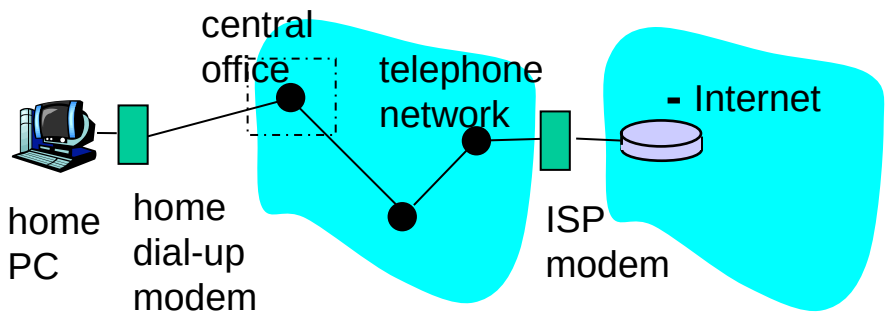


Source: Jim Kurose, Keith Ross: Computer Networking: A Top Down Approach - 5th edition, Addison-Wesley, April 2009.

- **End Systems (hosts):**
  - run application programs
  - e.g. Web, email
  - at “edge of network”
- **Communication Services** provided to apps:
  - reliable data delivery from source to destination
  - “best effort” (unreliable) data delivery
- **Client/Server Model**
  - client host requests and receives service from always-on server
  - e.g. Web browser/server; email client/server
- **Peer-To-Peer (P2P) Model:**
  - minimal (or no) use of dedicated servers
  - e.g. Skype, BitTorrent

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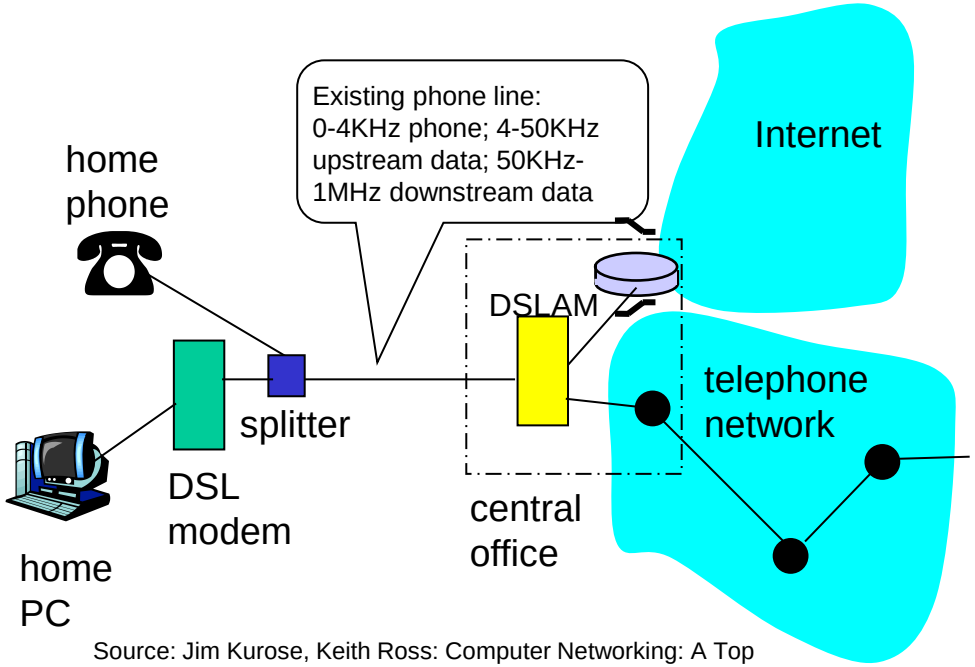
Dial-up Modem and Digital Subscriber Line (DSL)



Source: Jim Kurose, Keith Ross: Computer Networking: A Top Down Approach - 5th edition, Addison-Wesley, April 2009.

Dial-up Modem

- uses existing telephony infrastructure
  - home is connected to central office
- up to 56Kbps direct access to router
  - can’t surf and phone at same time: not “always on”



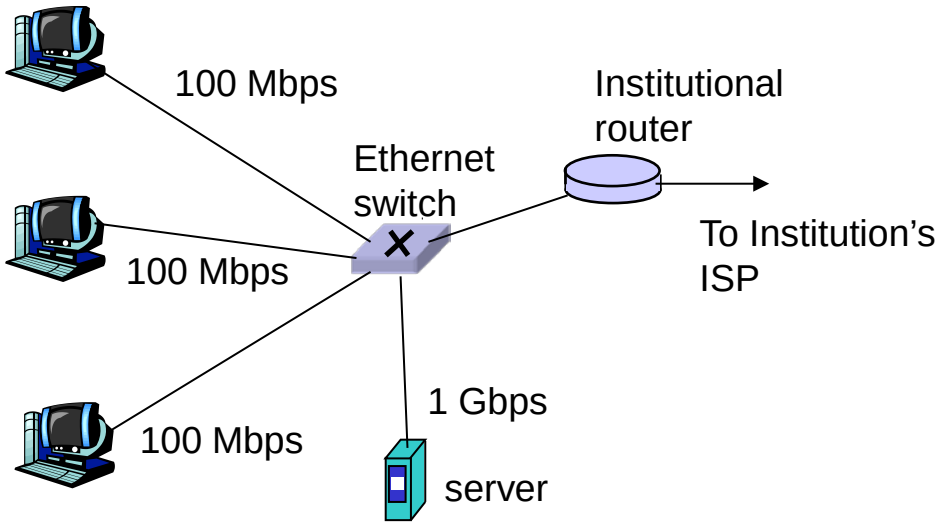
Source: Jim Kurose, Keith Ross: Computer Networking: A Top Down Approach - 5th edition, Addison-Wesley, April 2009.

Digital Subscriber Line (DSL)

- also uses existing telephone infrastructure
- up to 1 Mbps upstream
- up to 8 Mbps downstream
- … in some cases even higher data rates

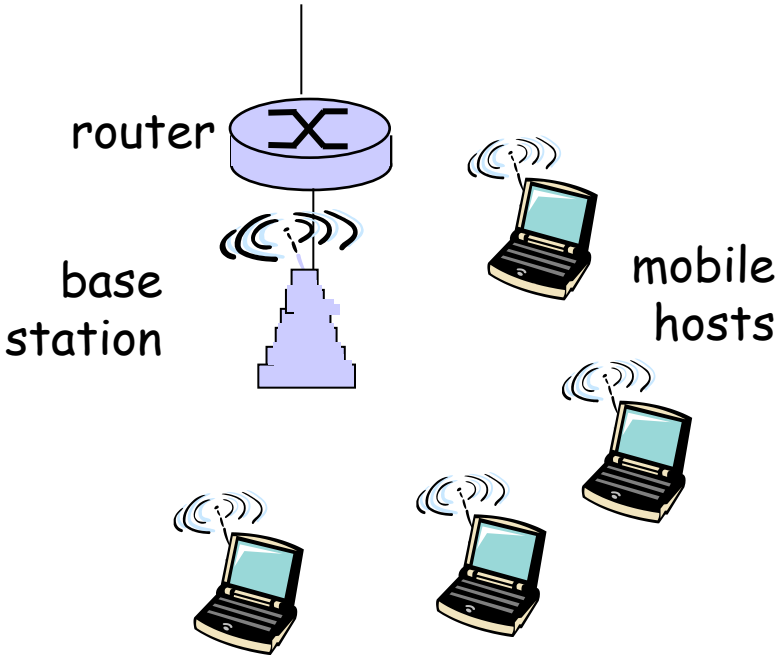
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Internet Access via Local Area and/or Wireless Networks



Local Area Networks

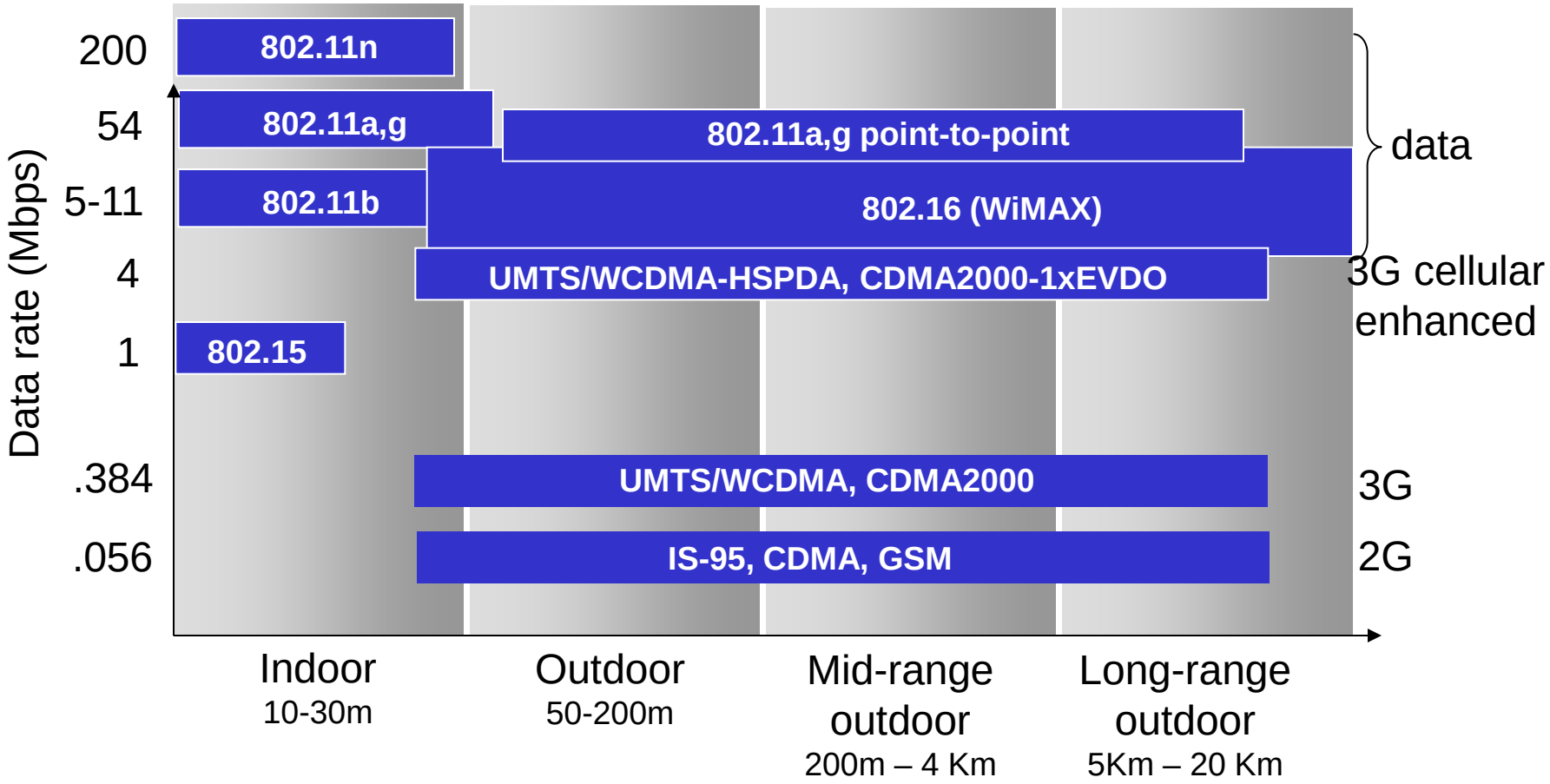
- typically used in companies, universities
- Used in many homes
- 10 Mbs, 100Mbps, 1Gbps, 10Gbps Ethernet



Wireless Access Networks

- shared wireless access network connects end system to router
- via base station aka "access point"

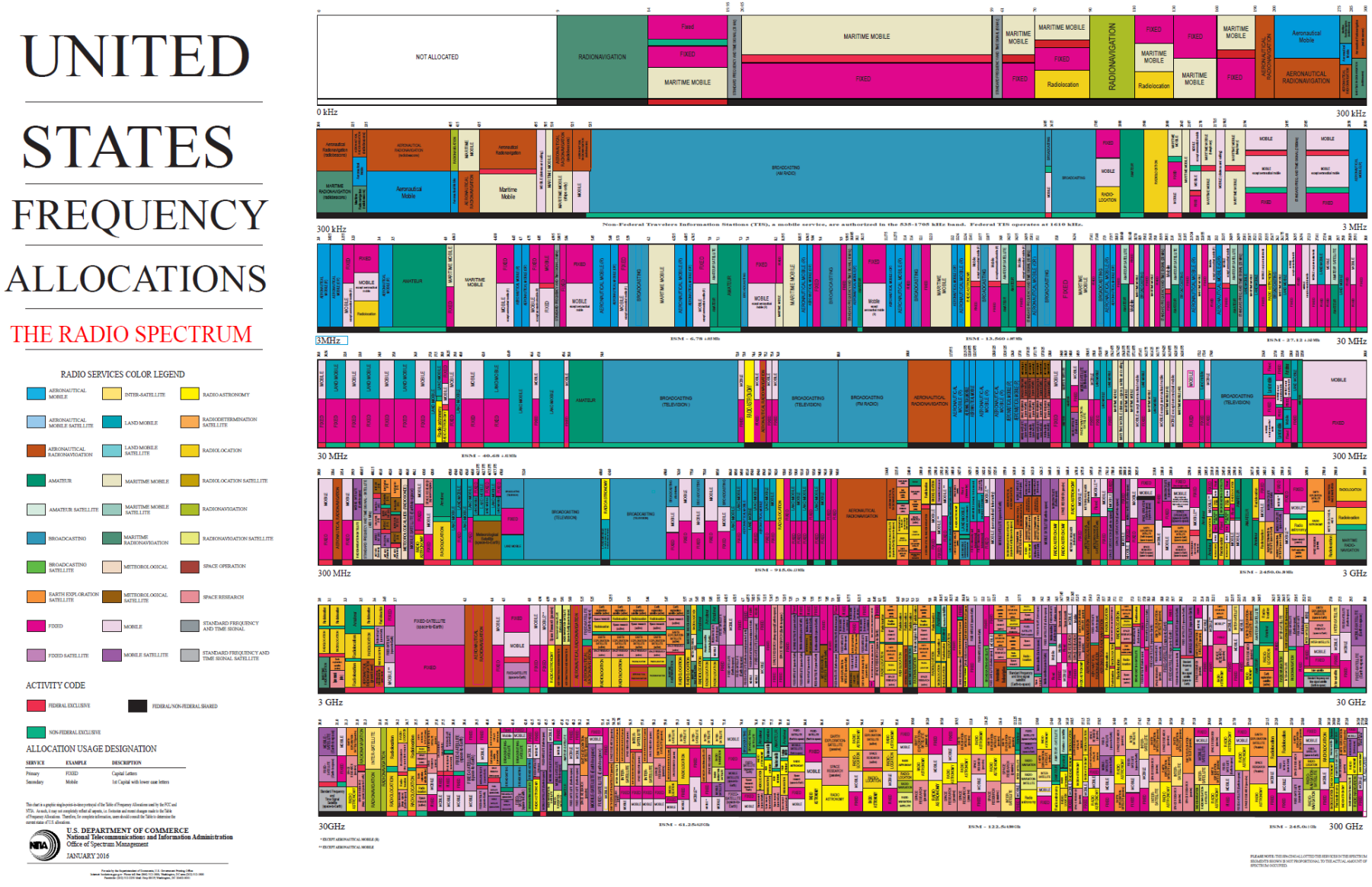
Characteristics of selected wireless link standards



Source: Jim Kurose, Keith Ross: Computer Networking: A Top Down Approach - 5th edition, Addison-Wesley, April 2009.

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Other Wireless Links – Spectrum Map – US



https://upload.wikimedia.org/wikipedia/commons/c/c7/United_States_Frequency_Allocations_Chart_2016_-_The_Radio_Spectrum.pdf

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Regulatory Authorities: Frequency Trading

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- Follow-up: Merger H3G/Orange
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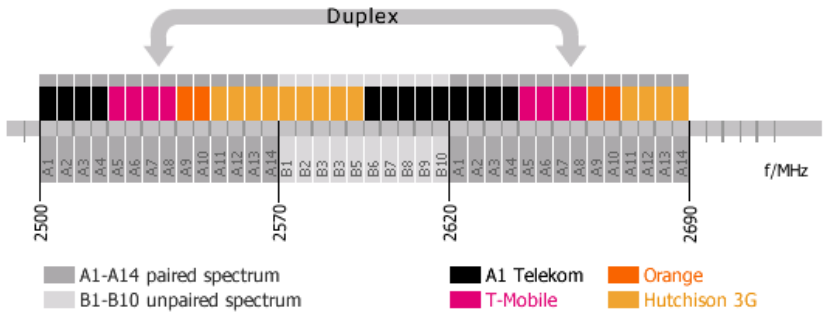
Home › Telecommunications › Frequency Allocation › Allocation procedures › 2600 MHz (2010) › Auction results

2600 MHz (2010) Auction results

The table below shows the final results of the 2,6 GHz auction.

Applicants	Paired Frequency Packages	Unpaired Frequency Packages	Euro
A1 Telekom Austria AG	4 (A1-A4, 2x20MHz)	5 (B6-B10, 25MHz)	13.248.223.-
T-Mobile Austria GmbH	4 (A5-A8, 2x20MHz)	-	11.247.323.-
Orange Austria Telecommunication GmbH	2 (A9-A10, 2x10MHz)	-	4.001.003.-
Hutchison 3G Austria GmbH	4 (A11-A14, 2x20MHz)	5 (B1-B5, 25MHz)	11.030.560.-
Total:	**14 (2x70MHz)**	**10 (50MHz)**	**39.527.109.-**

This leads to the following assignment:



Remark: The channels B1, B6 and B10 might be restricted according to the technical conditions (guard channels).

Top | Print | Recommend

Current messages

28.06.2013
Communications Report 2012

07.05.2013
RTR-Position Paper on Net Neutrality

23.04.2013
Follow-up: Merger H3G/Orange

Further messages

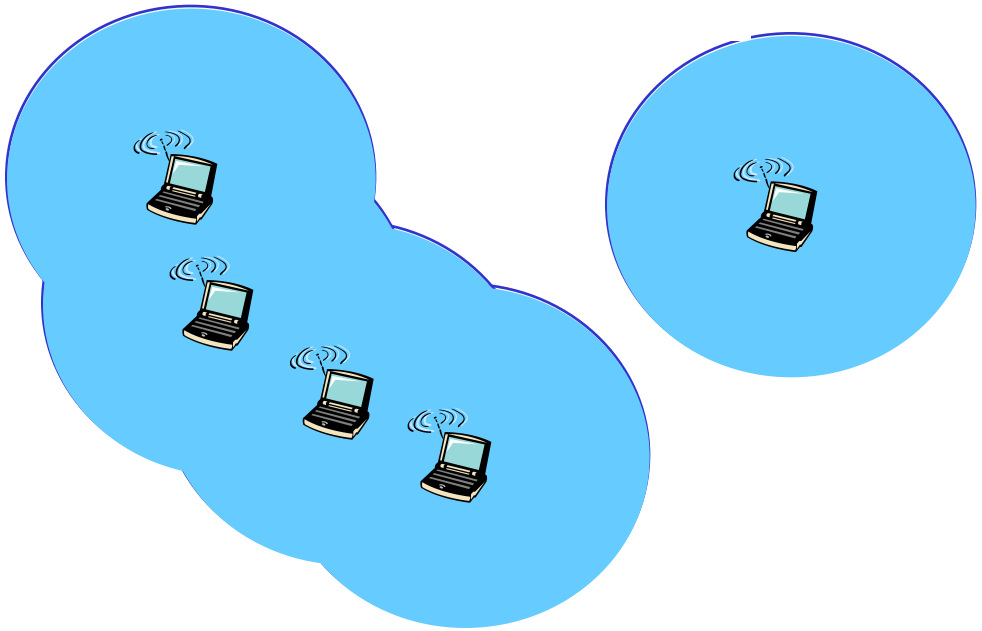
Copyright © Rundfunk und Telekom Regulierungs-GmbH (RTR-GmbH)

Publishing information

Source: https://www.rtr.at/de/tk/FRQ_2600MHz_2010_AE

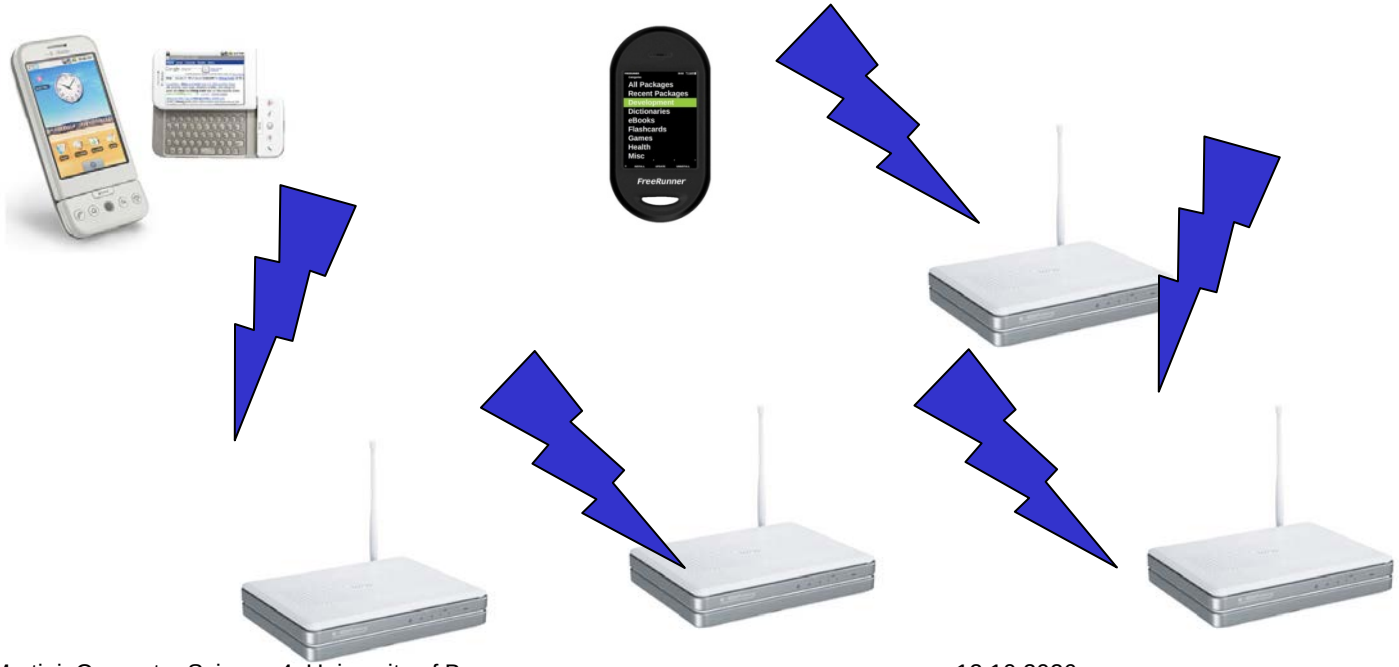
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Alternative Ad-hoc mode / Mesh networks



ad hoc mode

- no base stations
- nodes can only transmit to other nodes within link coverage
- nodes organize themselves into a network: route among themselves



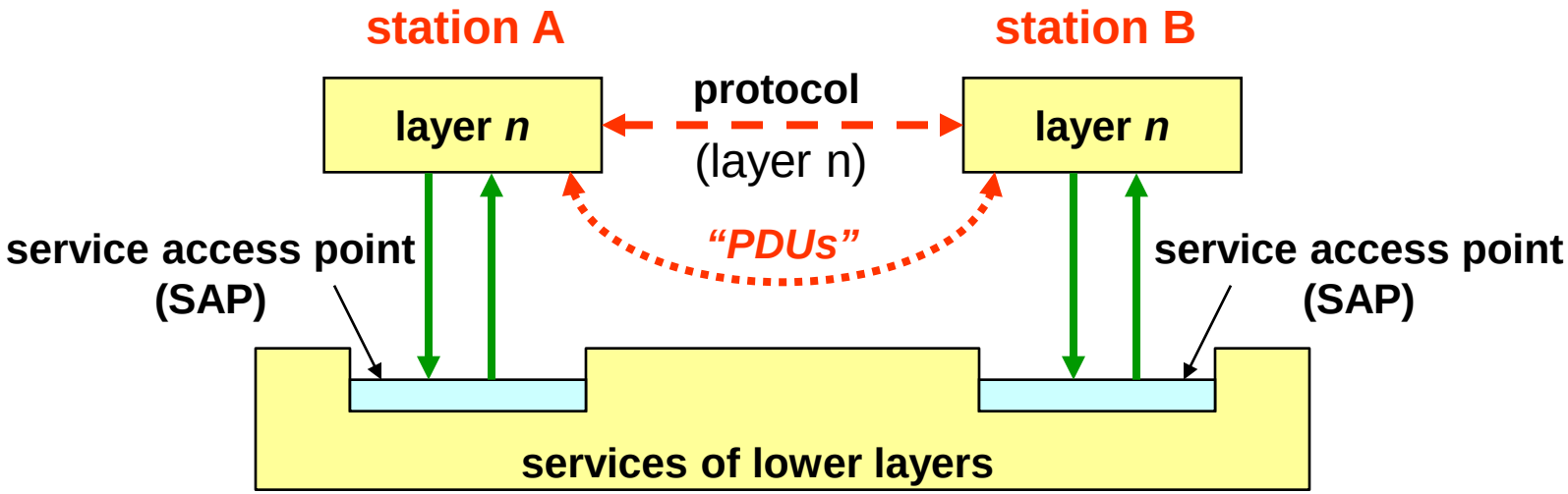
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1.4. Hierarchy Principle and Layer Models

Design, specification and standardization of communication mechanisms usually follow a hierarchical structure:

According to the hierarchy principle
each layer **takes care of specific problems**. (→ protocol of this layer)
layer *n* **directly communicates with layer *n* + *1*** (offers service for layer *n* + *1*)
layer *n* **directly communicates with layer *n* - *1*** (uses service from layer *n* - *1*)

A layer *n*
communicates **indirectly with the layer *n* of the peer instance**
exchanges **data units of a declared format** with the peer (Protocol data units "PDUs").



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The ISO Reference Model for Open Systems

The **International Standards Organization (ISO)**

has standardized the ISO reference model for open systems for the standardization of protocols:

Reference Model for Open Systems Interconnection (OSI)

Goals of the OSI model:

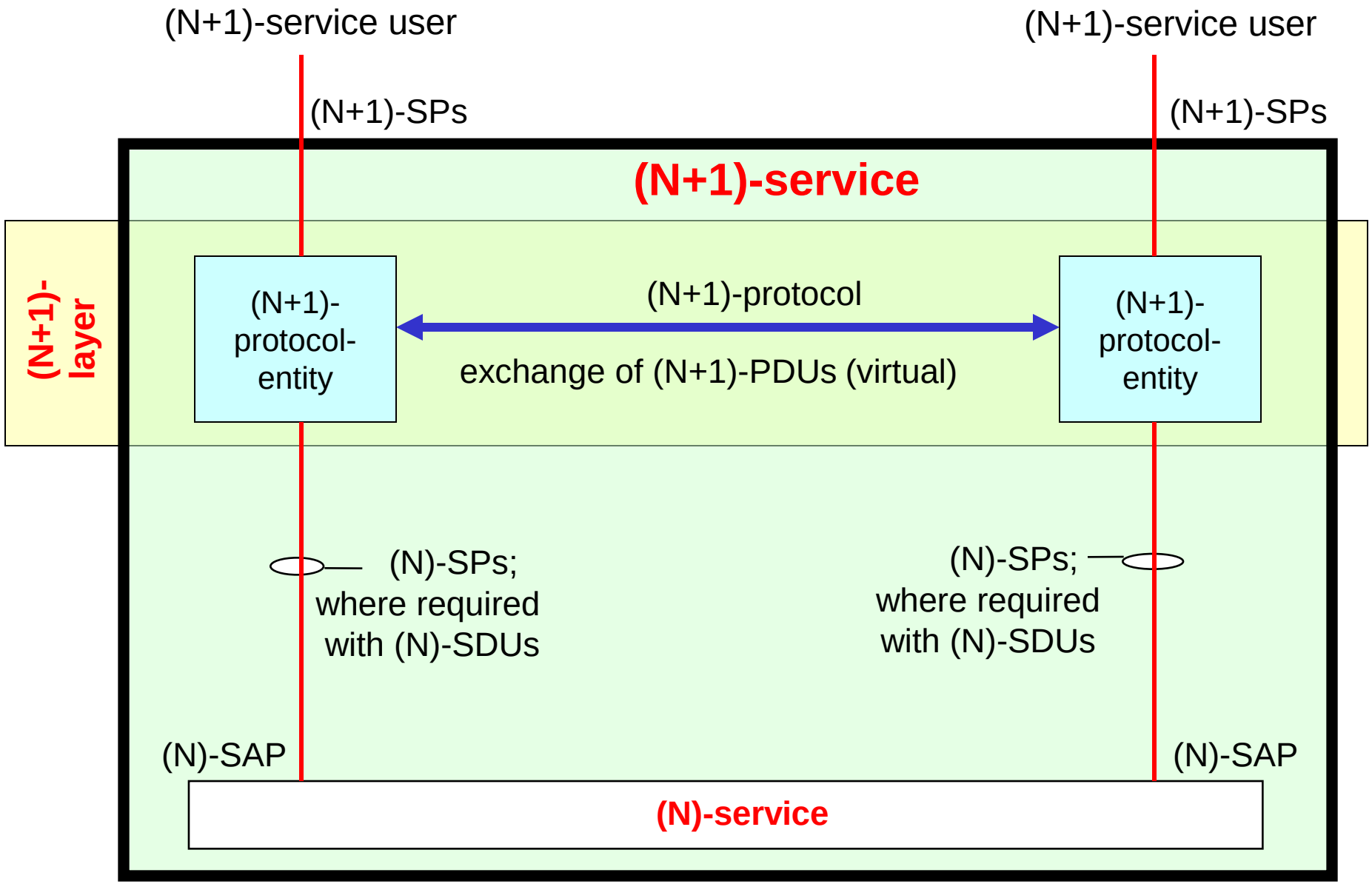
- standardized **nomenclature,**
- **structuring,**
- **framework for standards** (of protocols).

Remarks:

- The OSI model **does not standardize protocols**. It is only a framework for the standardization.
- The OSI -model **does not define an implementation.**
- Protocols within the scope of the OSI model **are defined independently from an implementation.**

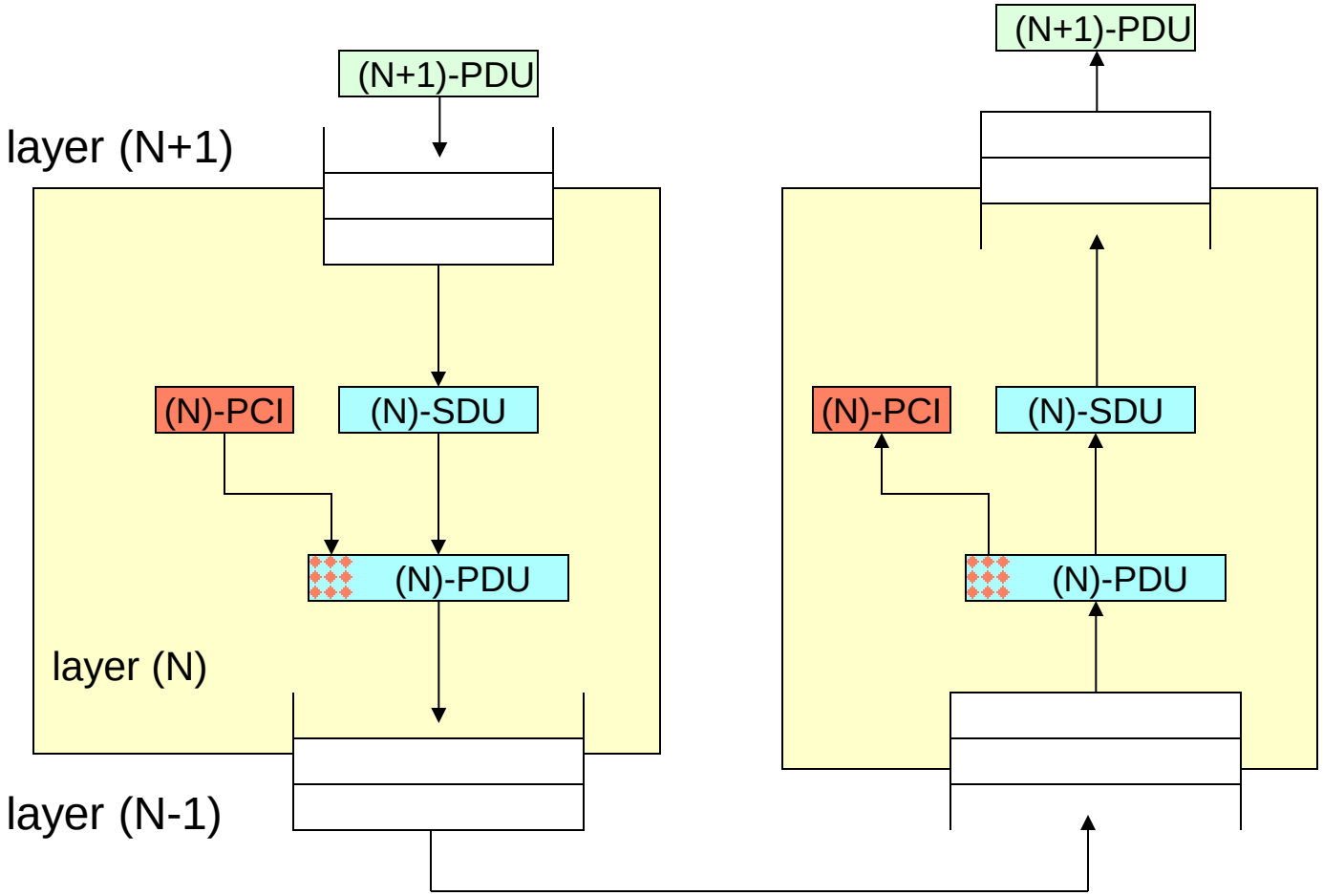
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(N+1)-Service and (N+1)-Protocol



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Simplified Model of a Communication System

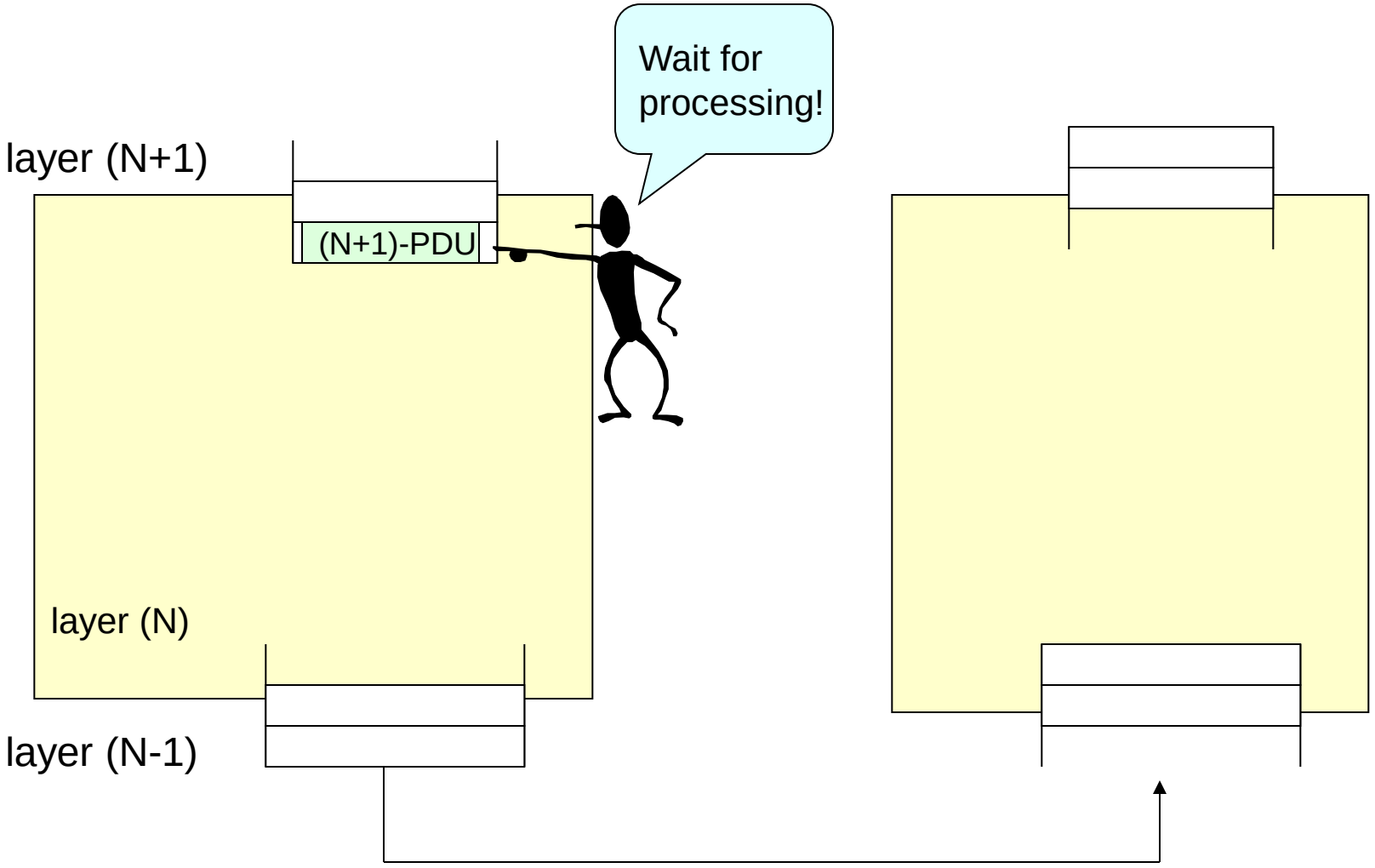


Important note:

An implementation with queues between the layers is neither mandatory nor desirable. The OSI model does not specify this kind of implementation detail.

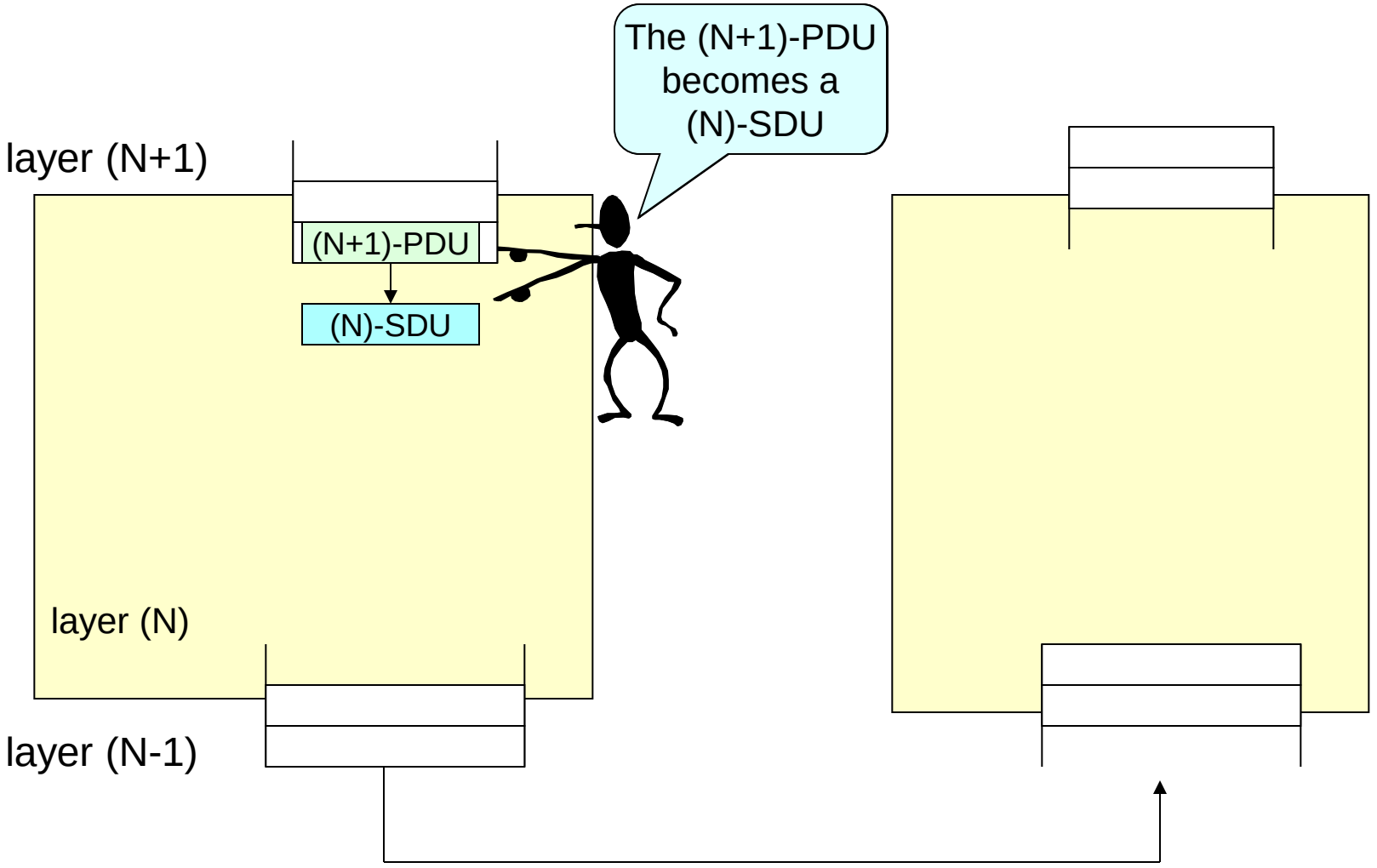
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Sending a (N+1)-PDU to layer (N)



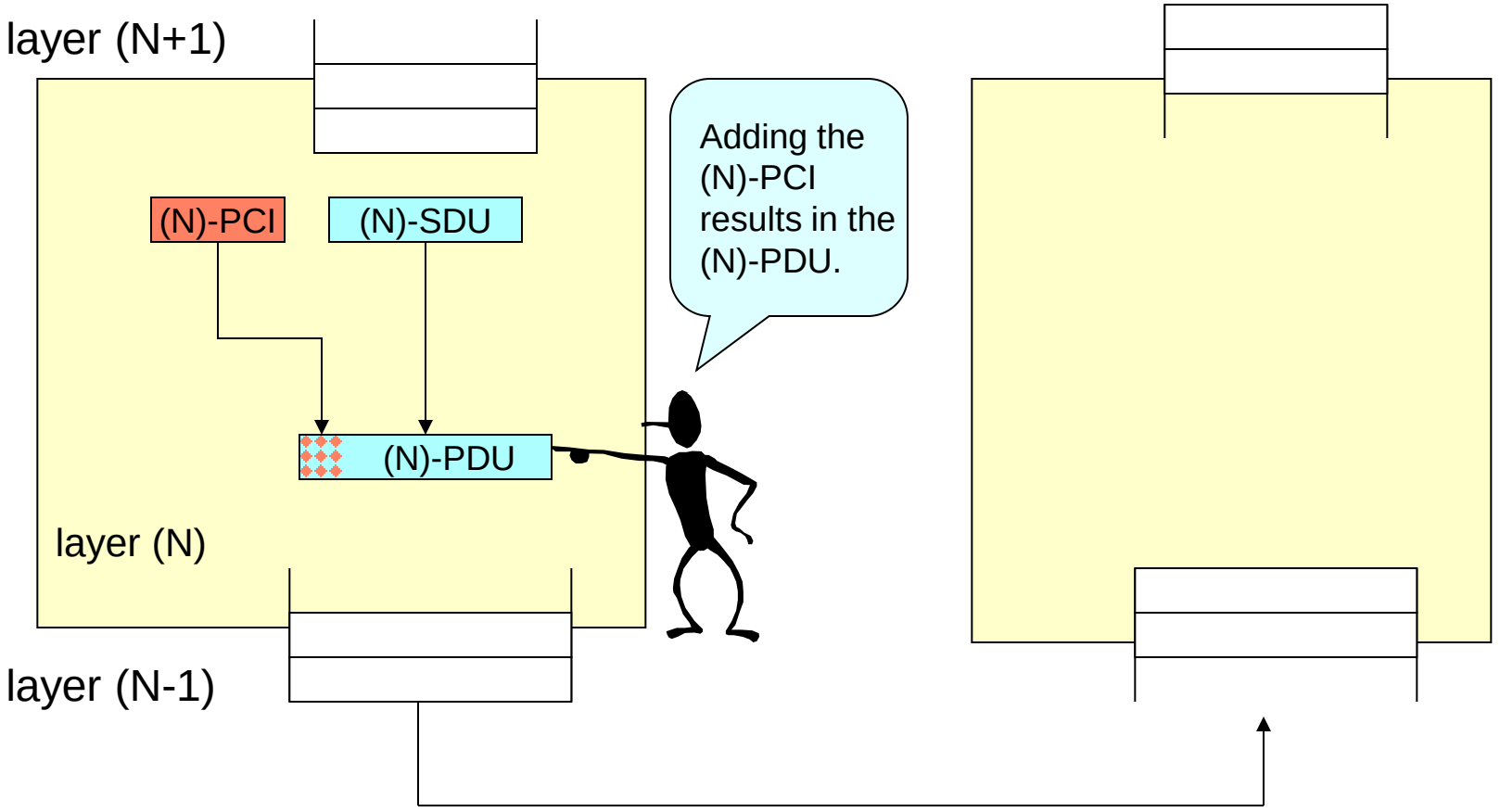
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A PDU becomes a SDU



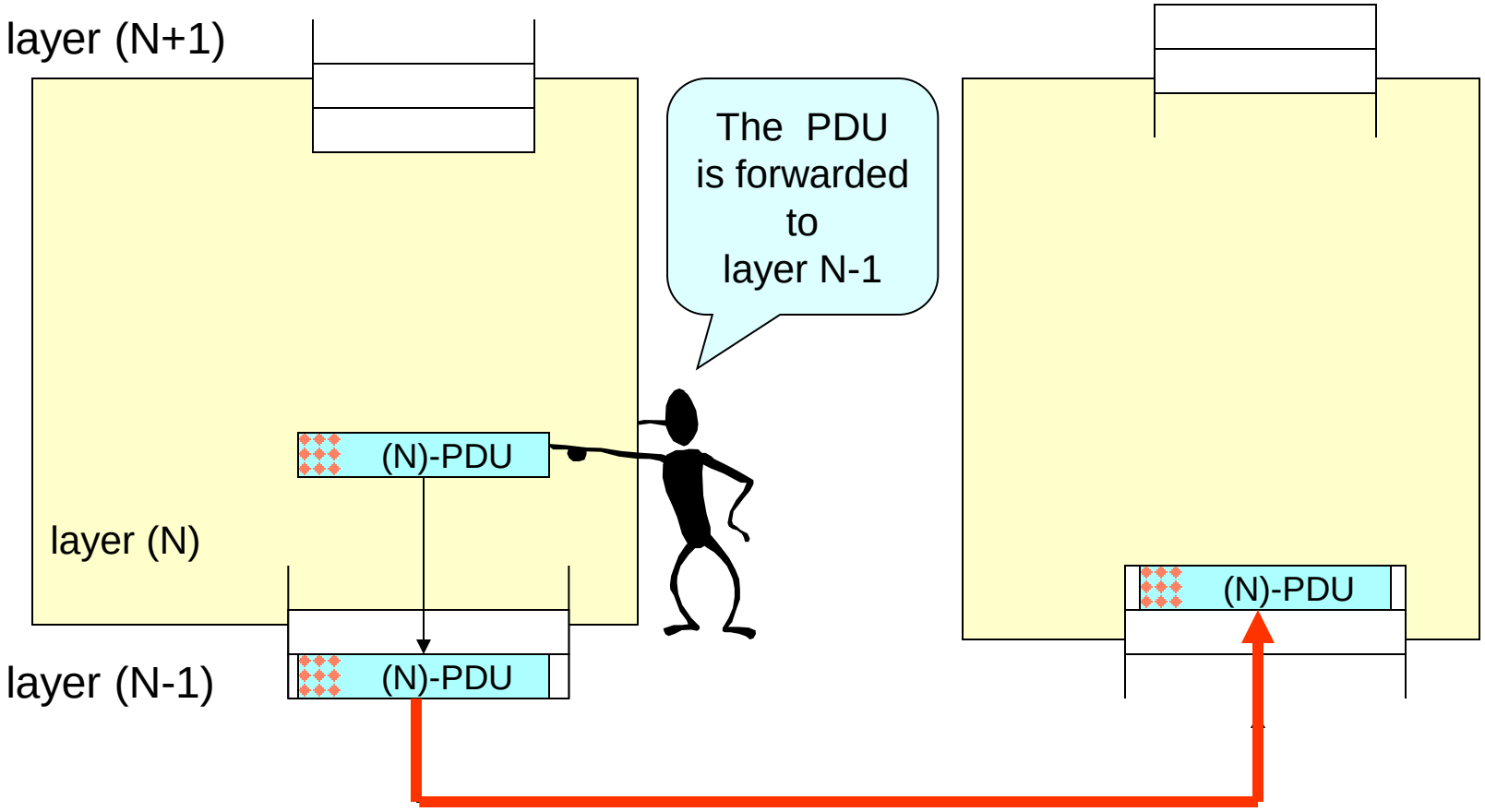
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Layer (N) forms a (N)-PDU from a (N)-SDU



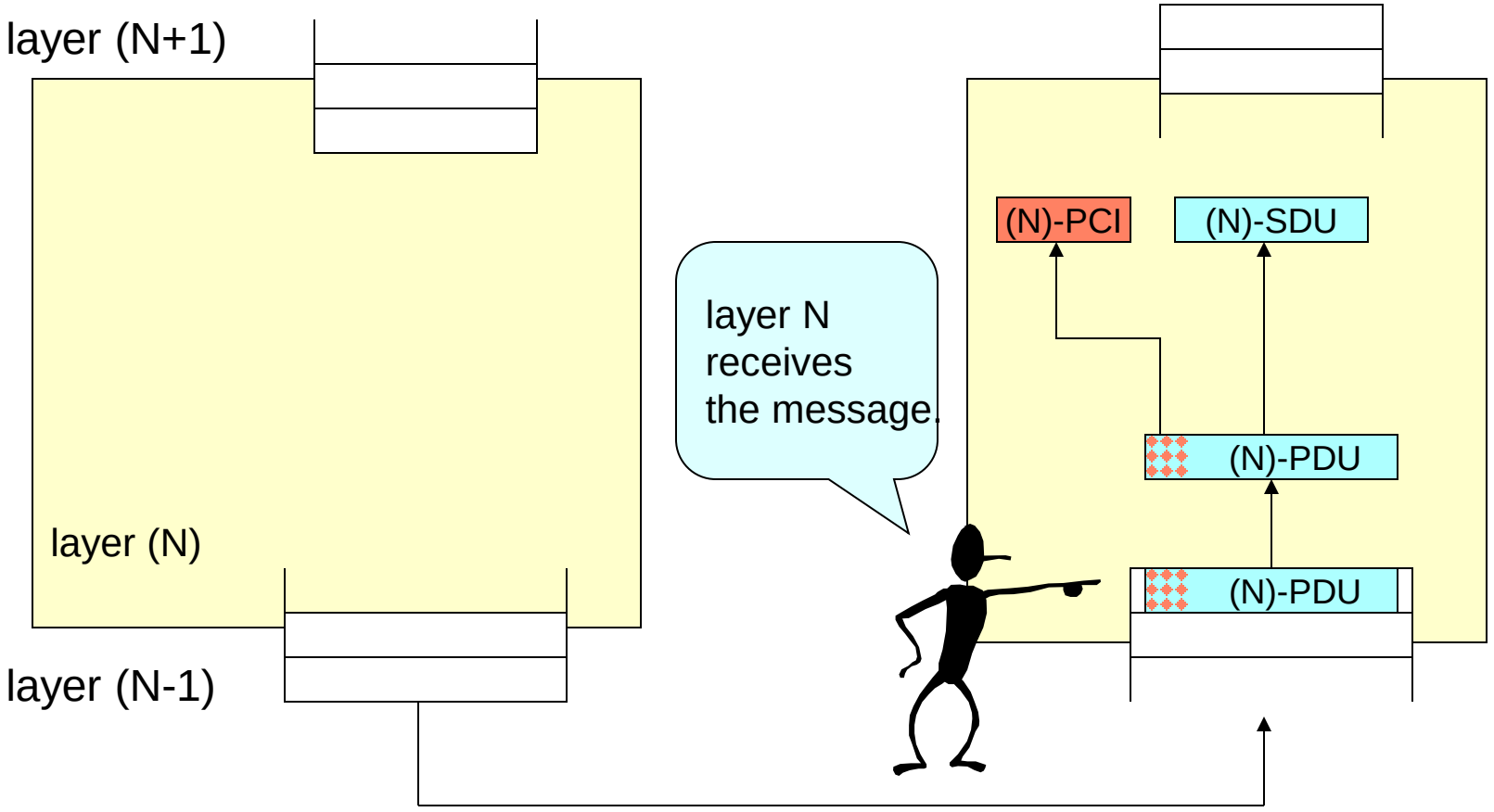
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Transfer to Layer (N-1) and Transmission



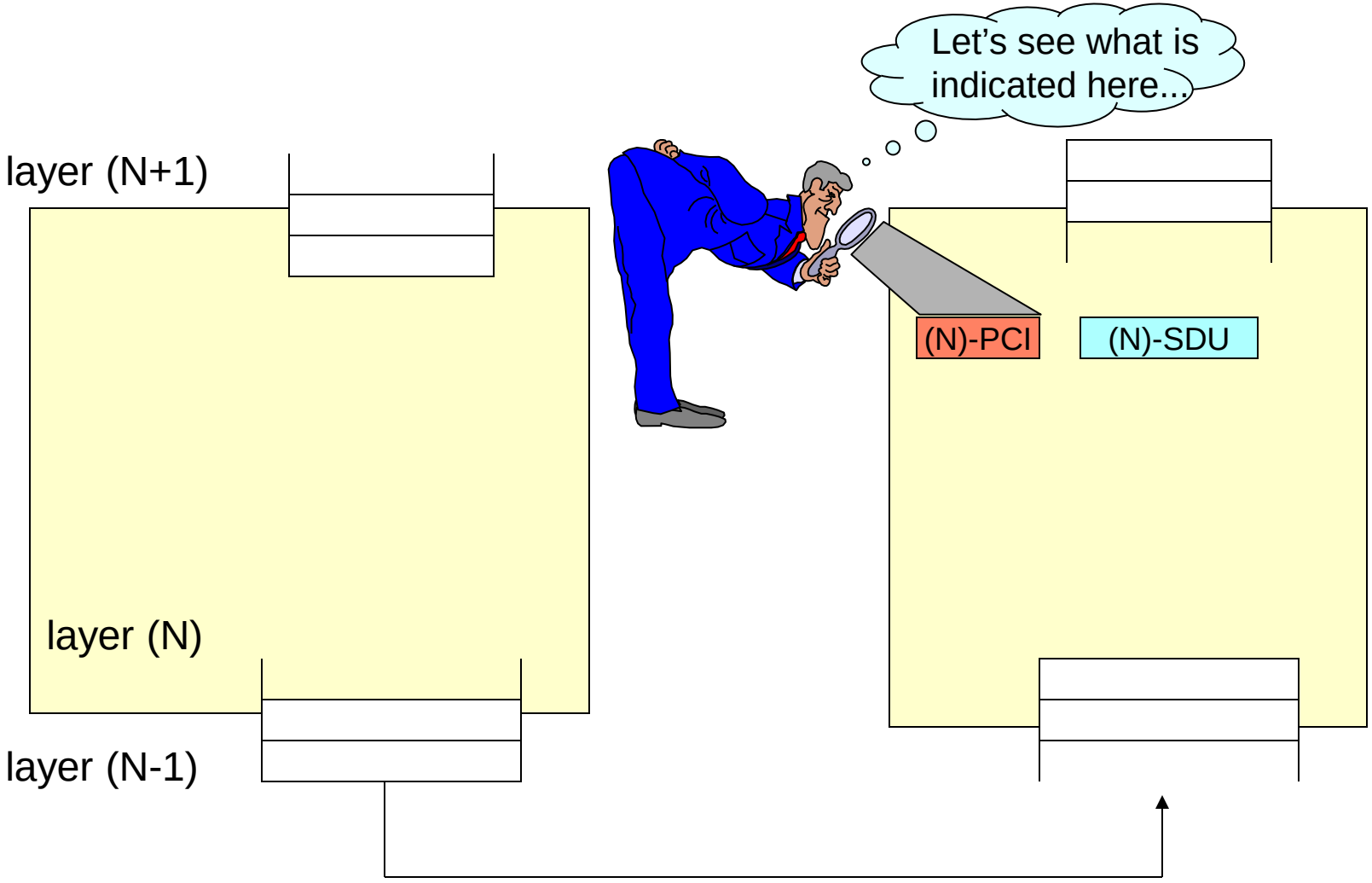
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Forward to Layer (N) and Process



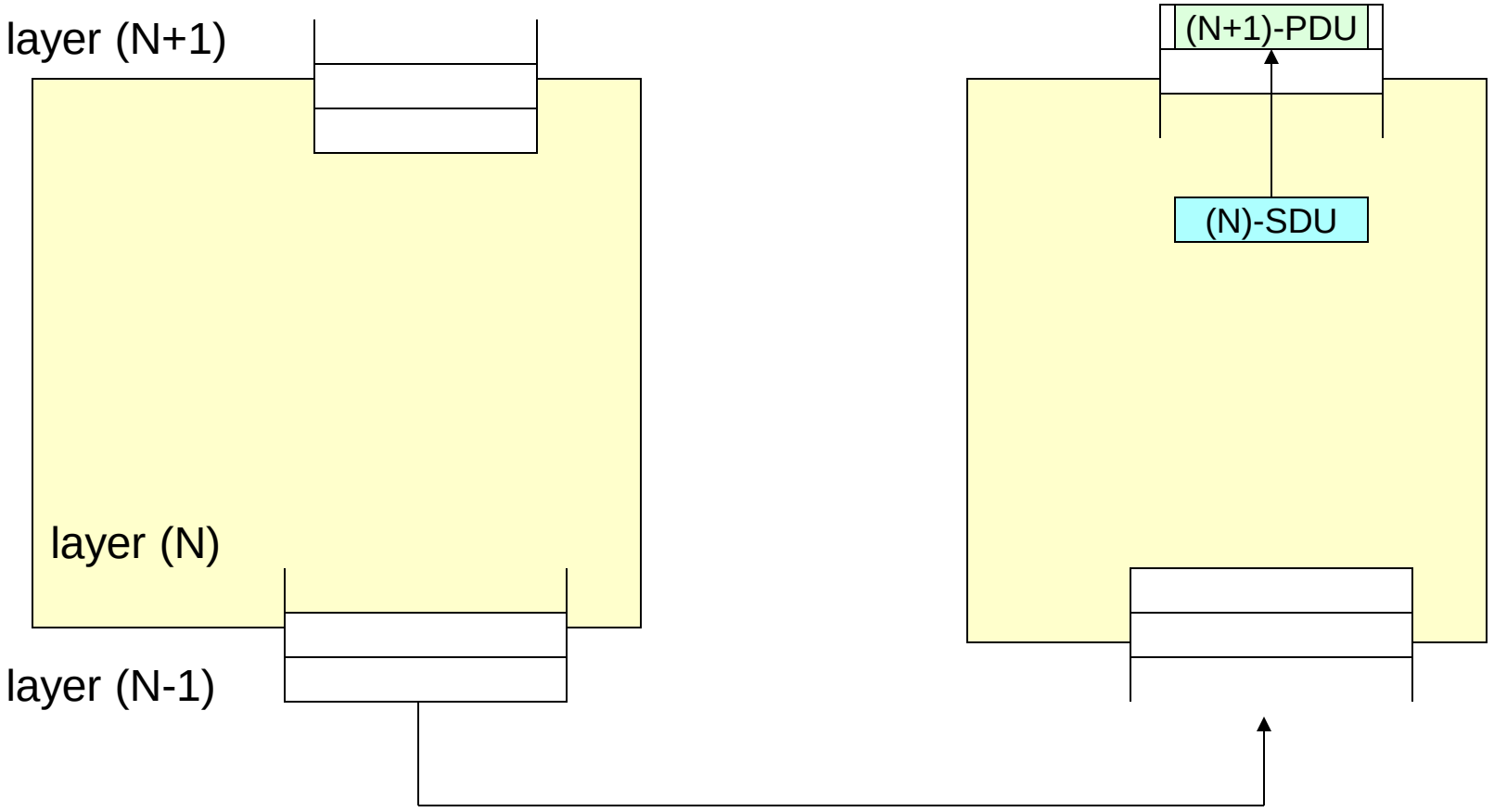
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Processing by Layer (N)



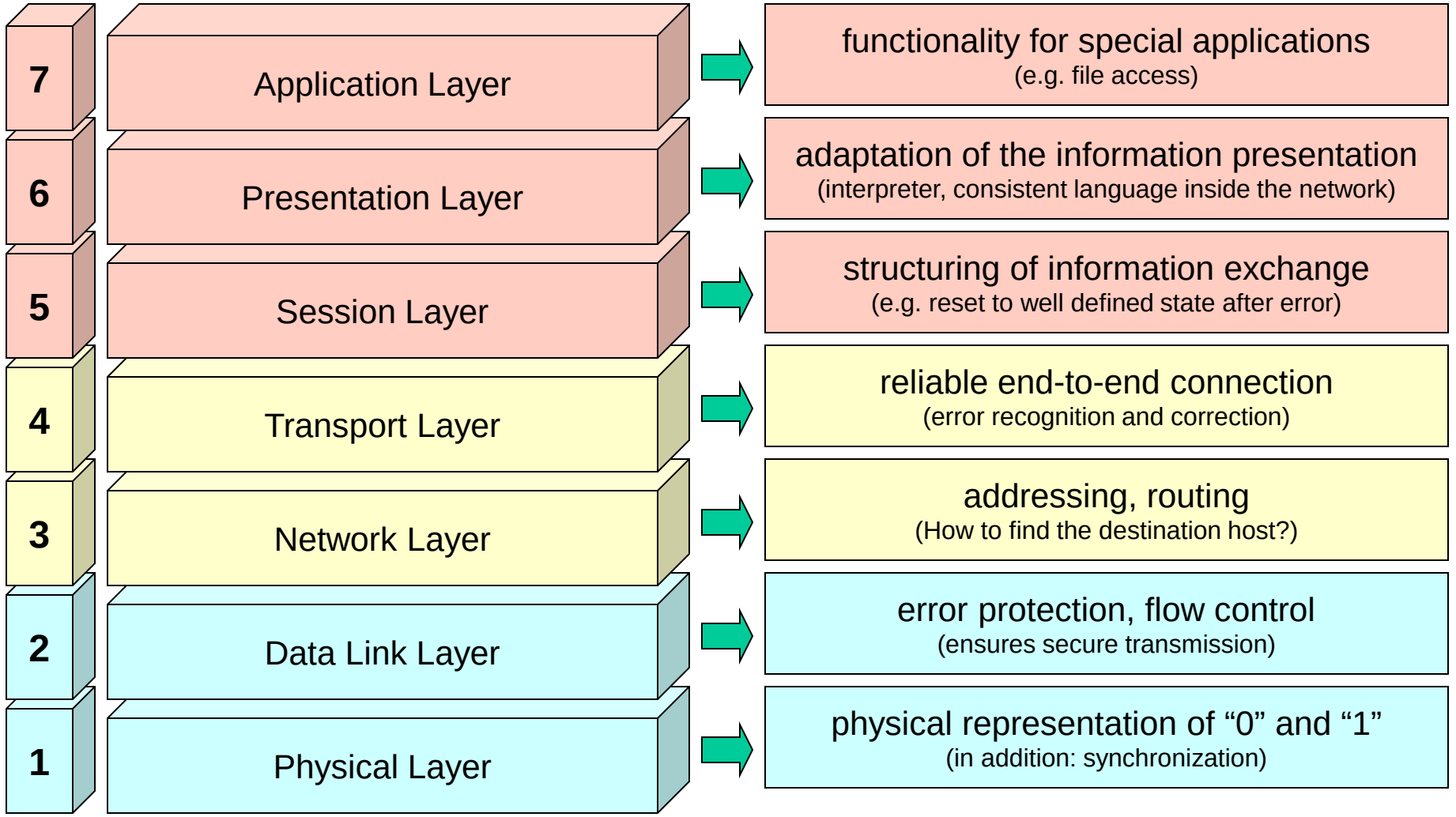
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Forward to Layer (N+1)



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The Seven Layers of the ISO/OSI Reference Model



application oriented: layers 5 to 7

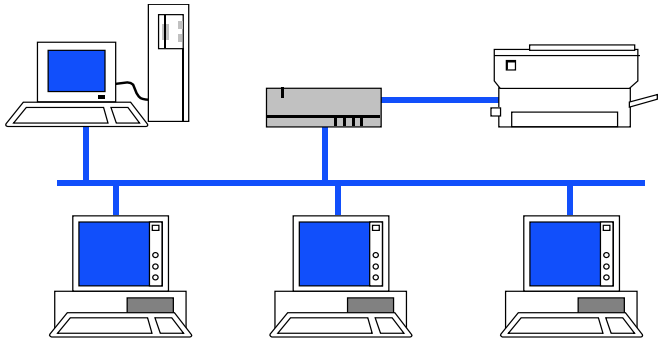
transport oriented: layers 3 and 4

technology oriented: layers 1 and 2

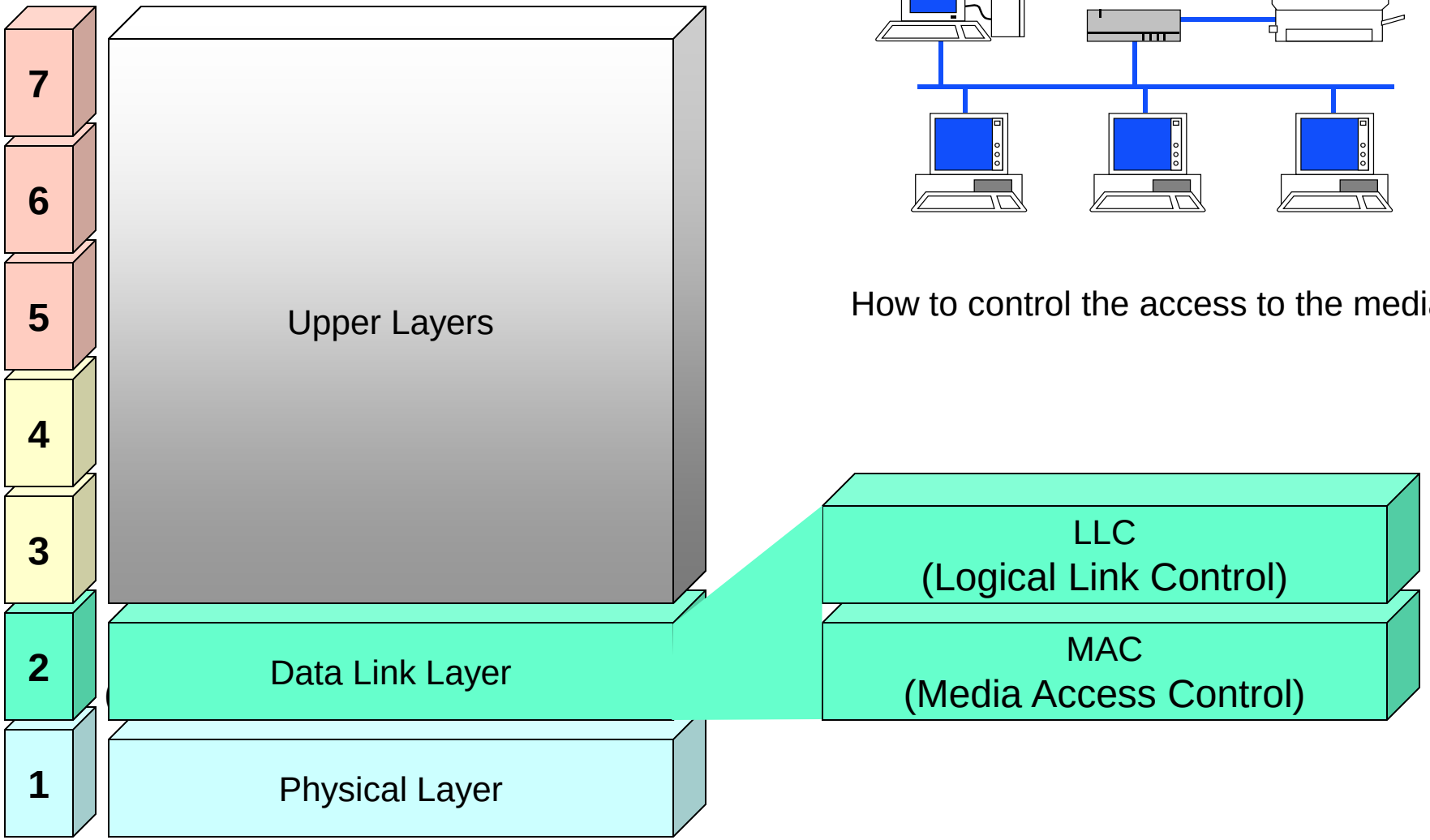
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The Layer Model of the IEEE LMSC (“IEEE 802”)

LAN/MAN Standards Committee der IEEE (“IEEE 802”)

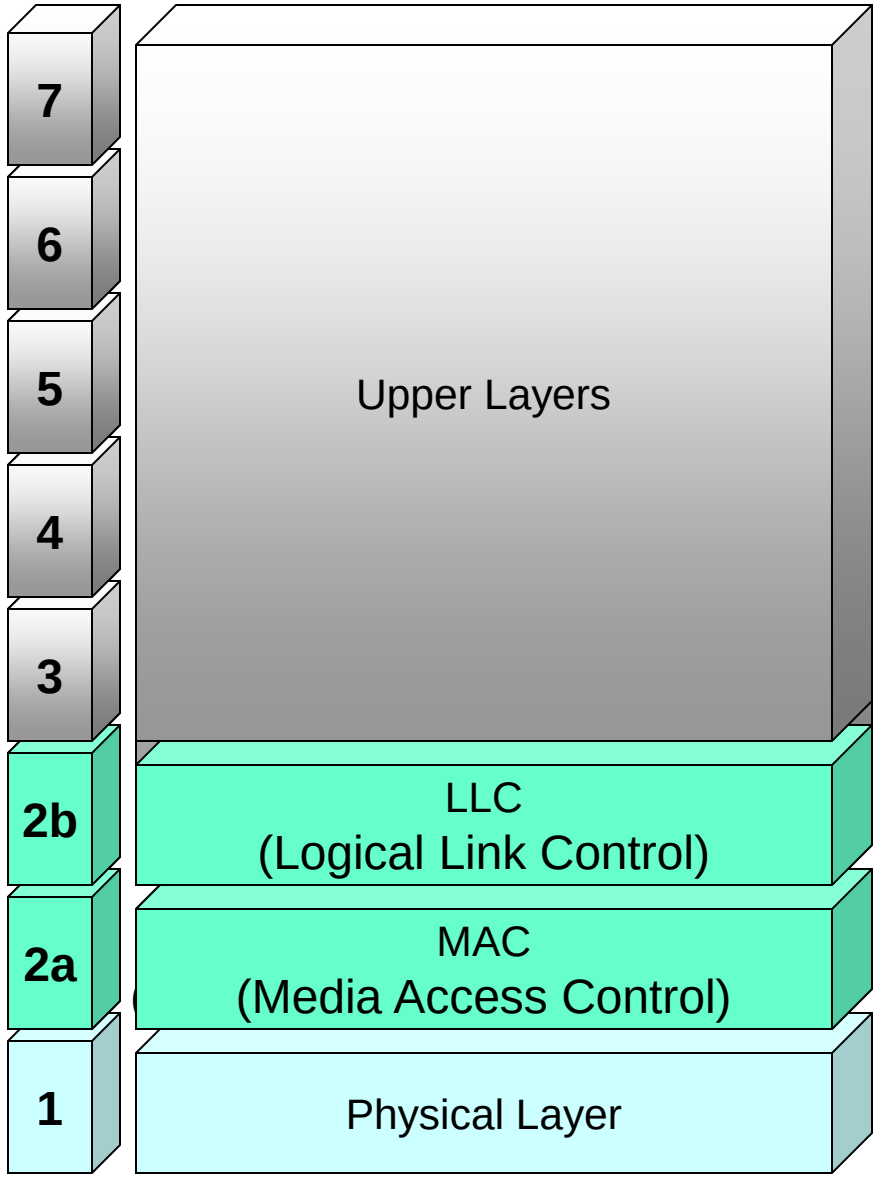


How to control the access to the media?



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The Layer Model of the IEEE LMSC ("IEEE 802") (2)



The LAN/MAN Standards Committee of the IEEE (better known as "IEEE 802") standardizes protocols for

- **layer 1,**
- **layer 2a and**
- **layer 2b**

of the OSI model.

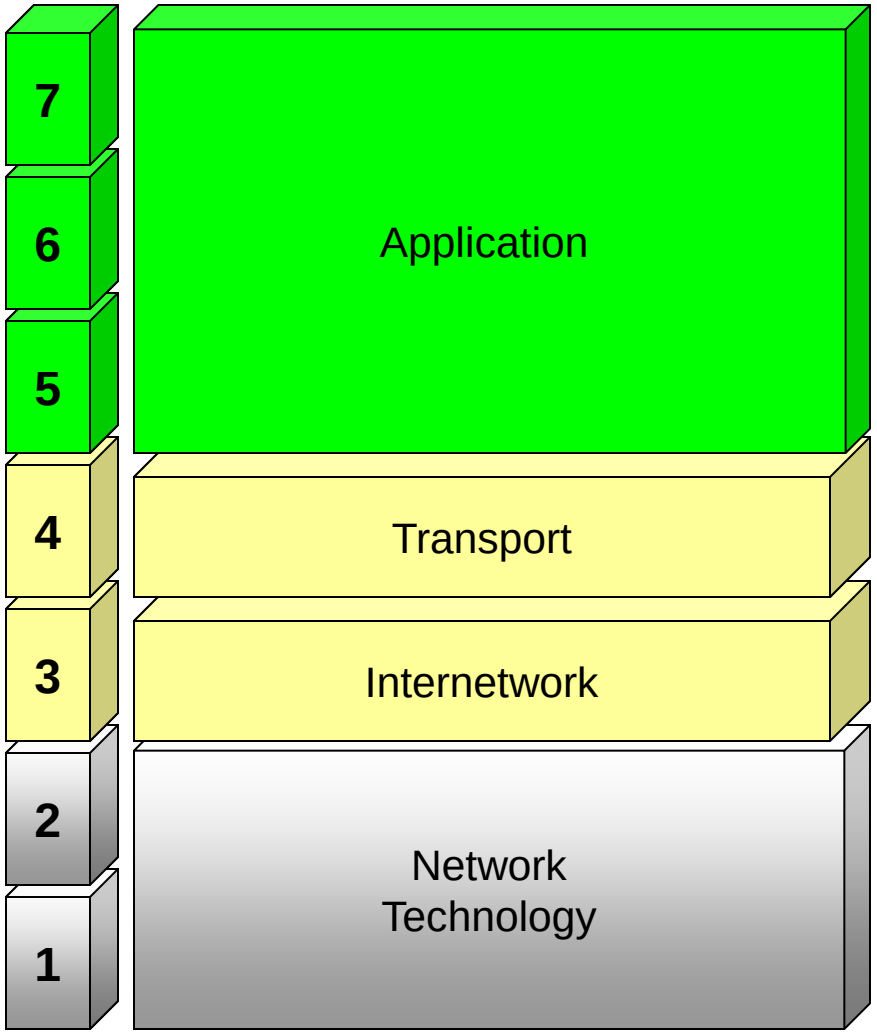
Well known protocols are

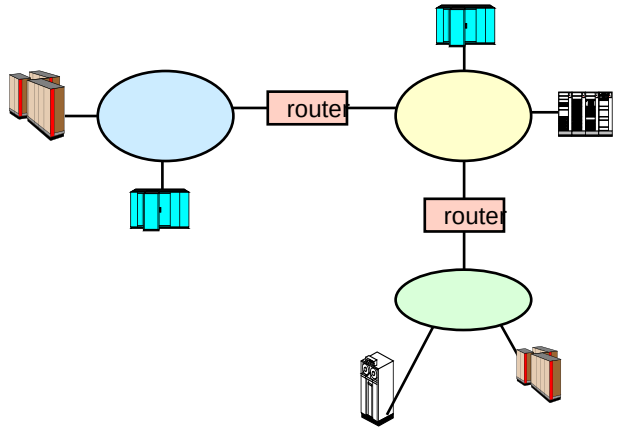
- "**Ethernet**" (IEEE 802.3),
- "**Token Ring**" (IEEE 802.5)
- "**Wireless LAN**" (IEEE 802.11)
- "**Wireless PAN**", "Bluetooth" (IEEE 802.15)

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The Layer Model of the IETF

Internet Engineering Task Force





The IETF “standardizes” protocols for

- **layer 3** and
- **layer 4**

of the OSI model.

Well known protocols are

- **“Internet Protocol”** (IP),
- **“Transmission Control Protocol”** (TCP) and
- **“User Data Protocol”** (UDP)

In addition, the IETF works on “applications” like e-mail, file transfer and remote login.

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1.5. Packet Sniffing with WireShark

WireShark is an **open source** tool for

- software and protocol development,
- troubleshooting,
- analysis,
- education,…



About Wireshark:
Wireshark is the world's foremost network protocol analyzer, and is the de facto (and often de jure) standard across many industries and educational institutions. Wireshark development thrives thanks to the contributions of networking experts across the globe. It is the continuation of a project that started in 1998.

(Source: http://www.wireshark.org/about.html, accessed Sept. 2013)

WireShark is not the only open source tool available for protocol analysis:

tcpdump / windump	(command line tool with textual output),
ngrep	(filters and displays network connections),
netstat	(command line tool to view open connections),
ettercap…	

A nice starting point for more tools is *http://sectools.org/*

Warning:
The Internet is governed by national and international laws!
Sniffing, hacking etc. may have **severe consequences** for the sniffer/hacker!



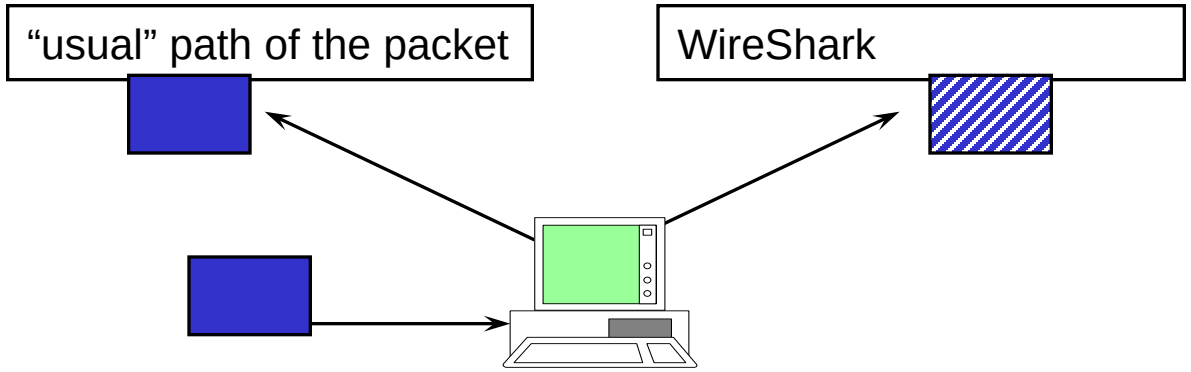
With our Security Lab (SecLab), our working group provides a playground for sniffing, hacking, spoofing, …
Interested? Be sure to attend our lecture "Network Security", held each summer semester.

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How WireShark Captures Packets

- WireShark uses a library to **capture packets** from interfaces.
  - Linux, FreeBSD,…: The library is called **libpcap**.
  - Windows (95 - XP): The library **WinPcap** is used.

- WireShark gets copies of the packets.

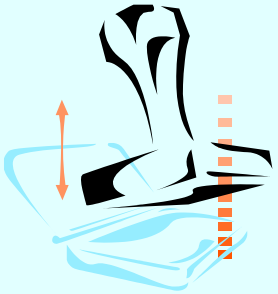


- WireShark identifies the protocol / packet type.
  - More than **600 protocols / packet types** are supported.
  - Identification is performed by modules called **“Dissectors”**.
  - WireShark is open source and **“User Specific Dissectors”** can be developed.

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A Word on Time Stamps of Packets

- WireShark displays **timing information** about the packets captured.
  - (Date &) time of day,
  - time since start of capture, or
  - time since previous packet.



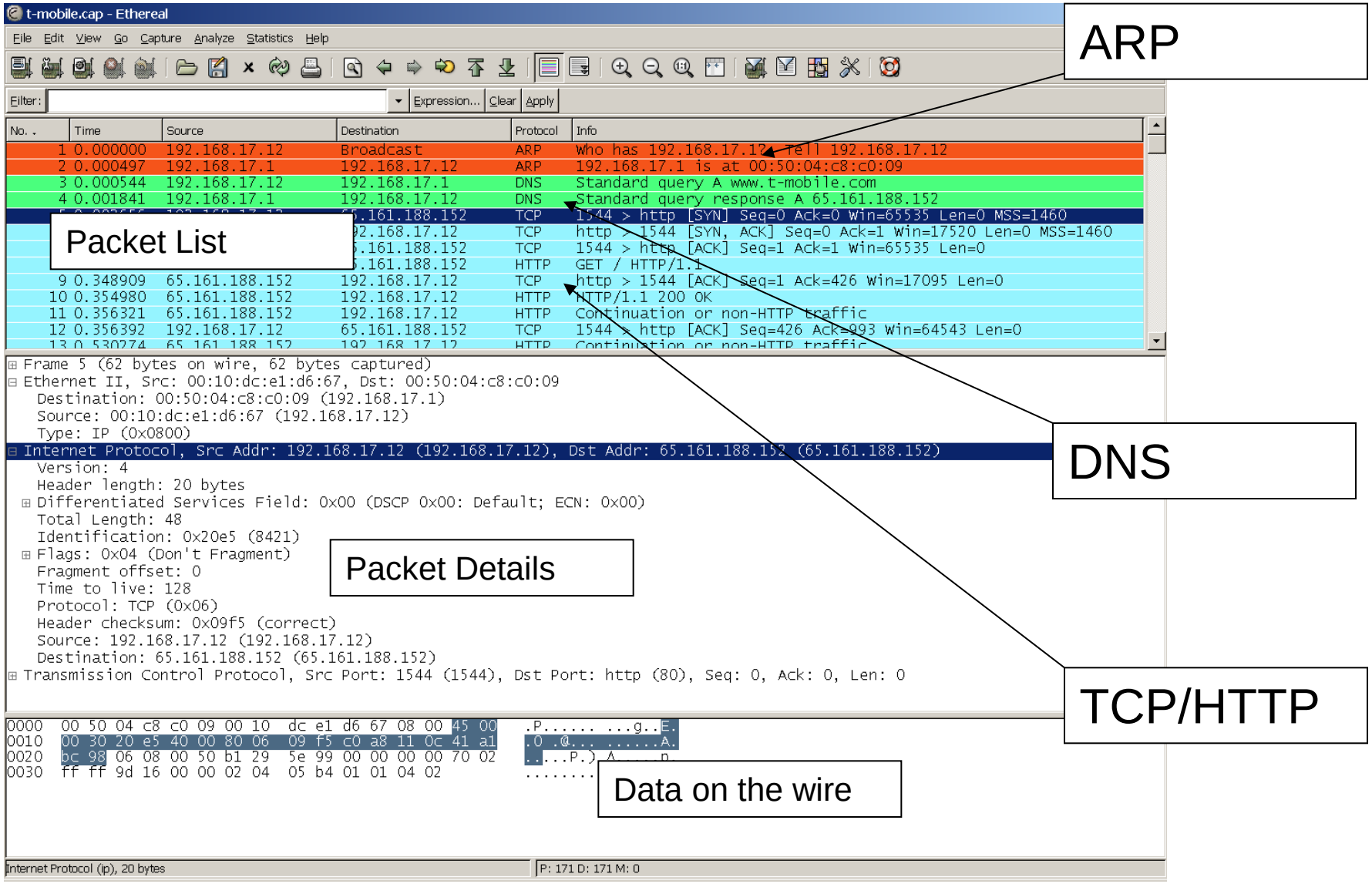
- The time stamp can be issued by (in order of **accuracy**)
  - the network card,
  - the operating system (i.e. network interface device driver), or
  - the packet capture library (libpcap, WinPcap).



- WireShark shows time in terms of ***µs***, but the **resolution** depends on
  - the hardware platform,
  - operating system,
  - network card behaviour, …

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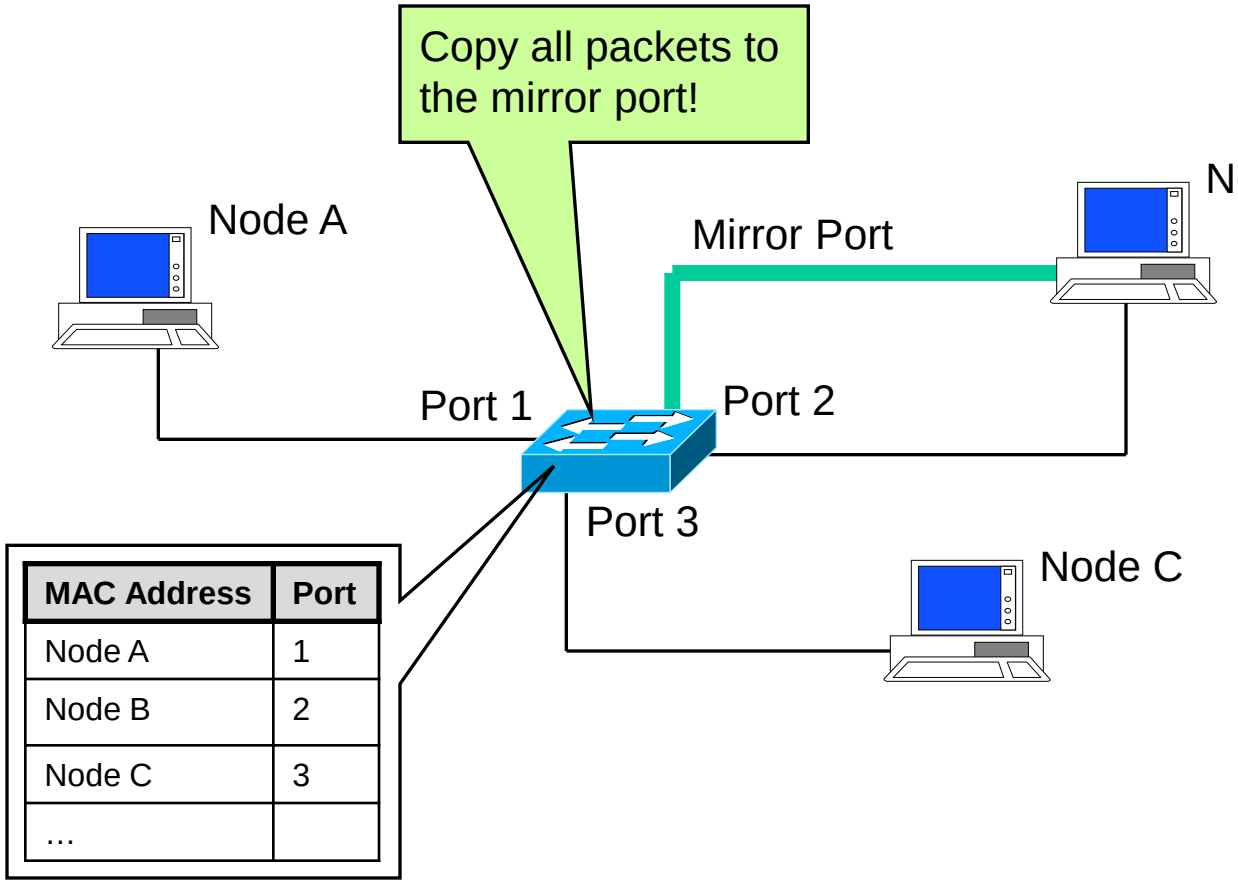
Getting Connected – WireShark Perspective



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Traffic on the LAN – What is “visible” to WireShark?

WireShark captures information units (packets/frames) sent by or to the host where WireShark is running.
A wider scope is possible when using “mirror ports” (only available on industry strength switches).



MAC Address	Port
Node A	1
Node B	2
Node C	3
…	



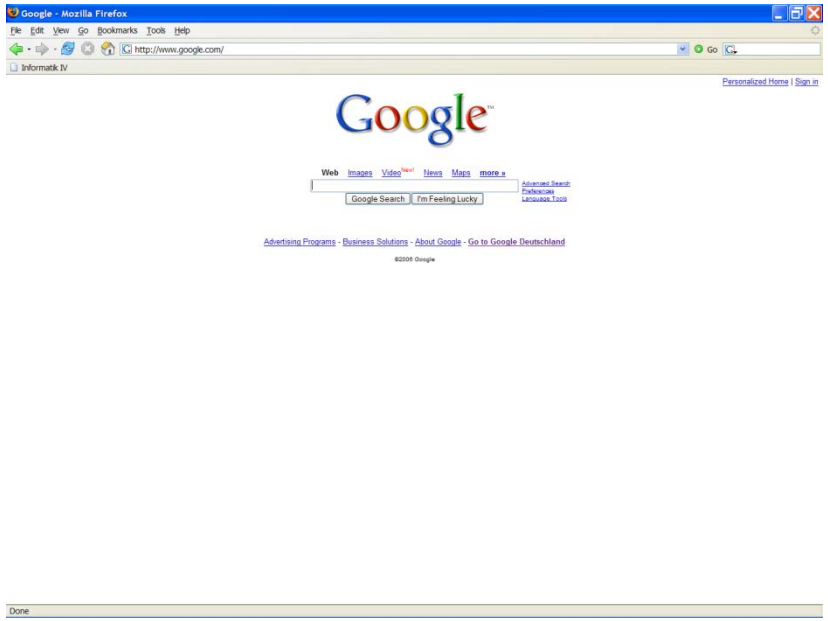
http://theithacan.org/blogs/brink/files/2007/12/130-126big-brother-is-watching-you-posters.jpg

Mirror ports receive the aggregated load. Is the line fast enough?

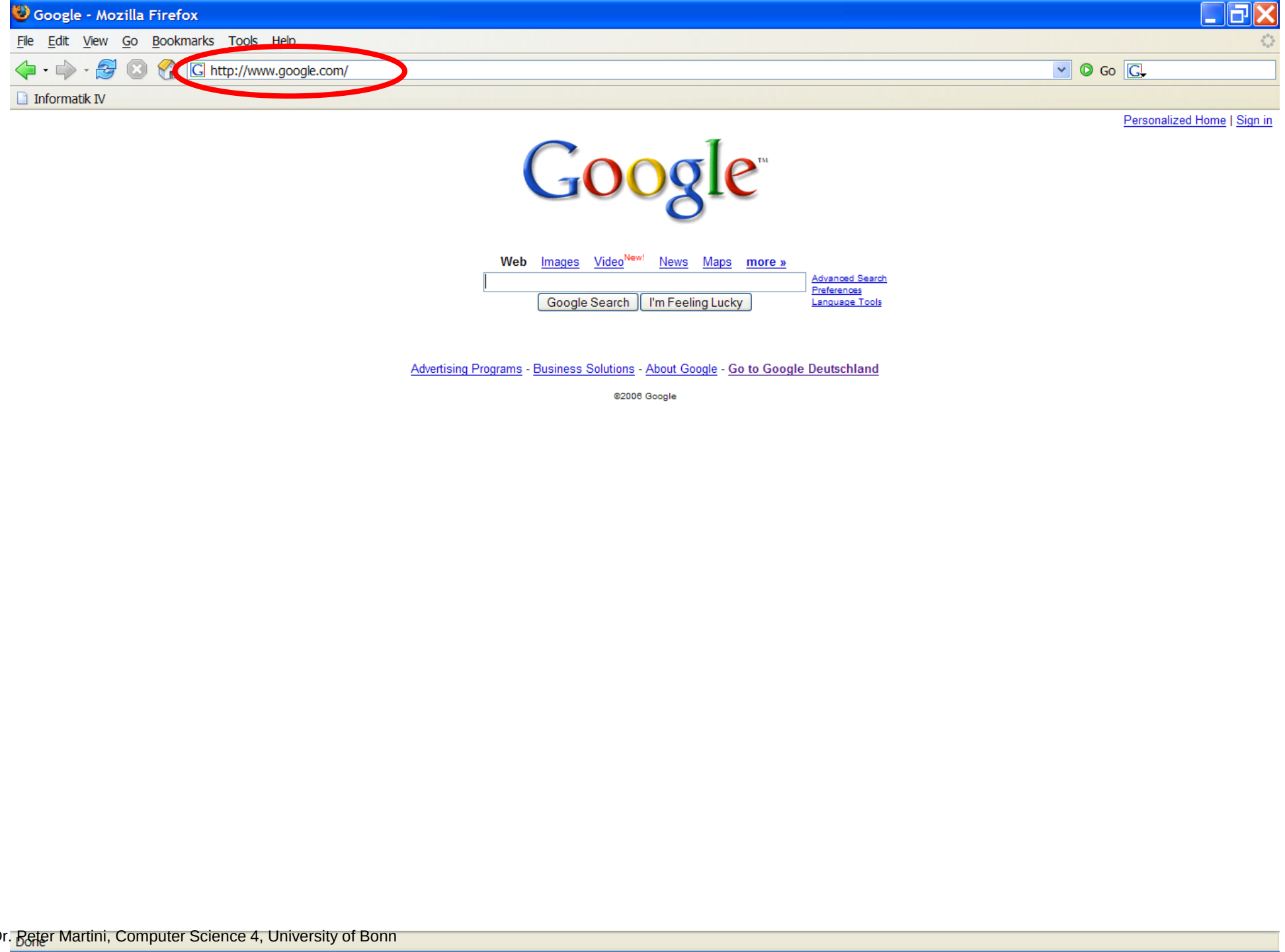
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Some Examples

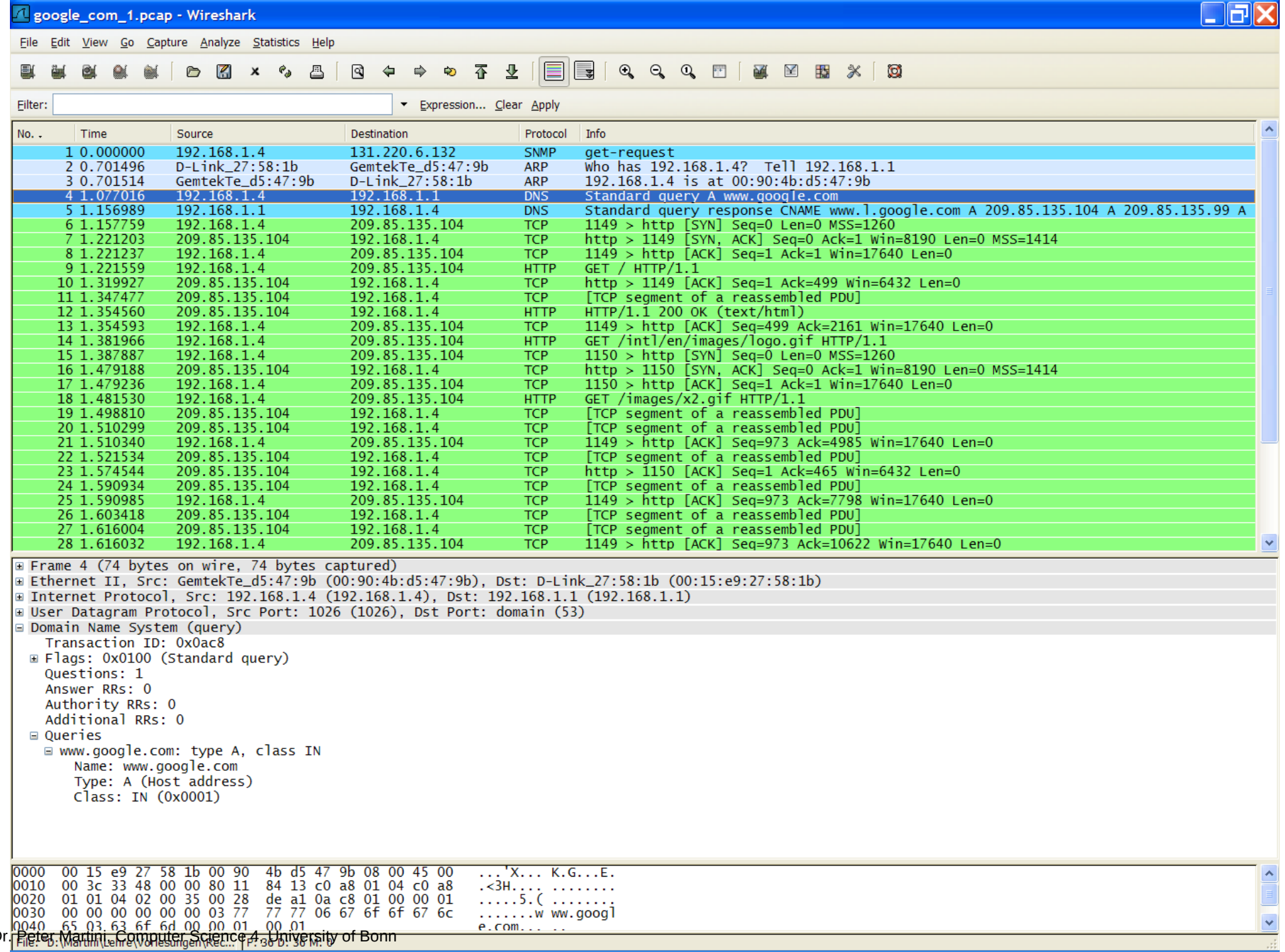
… presented on the following pages.



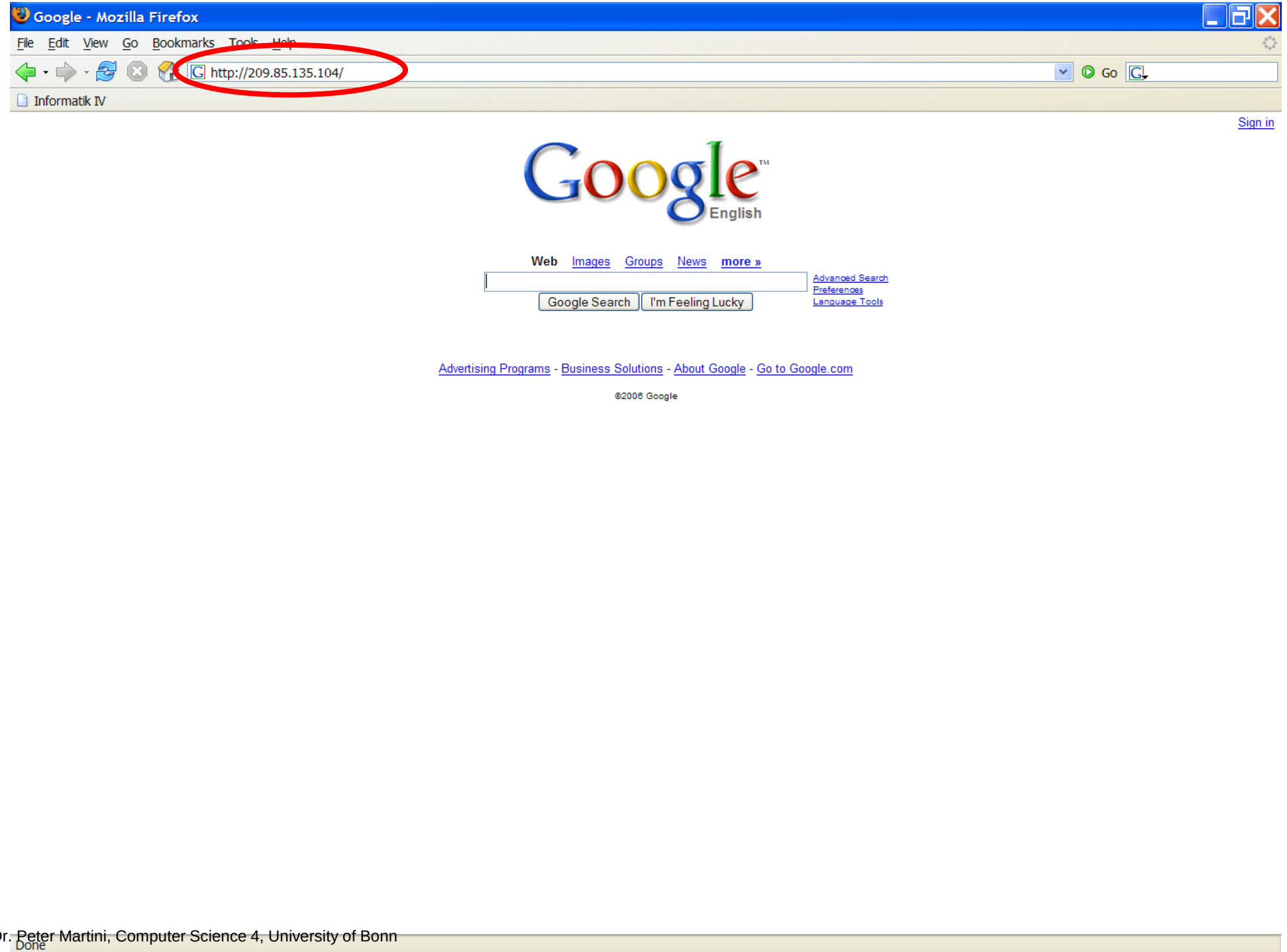




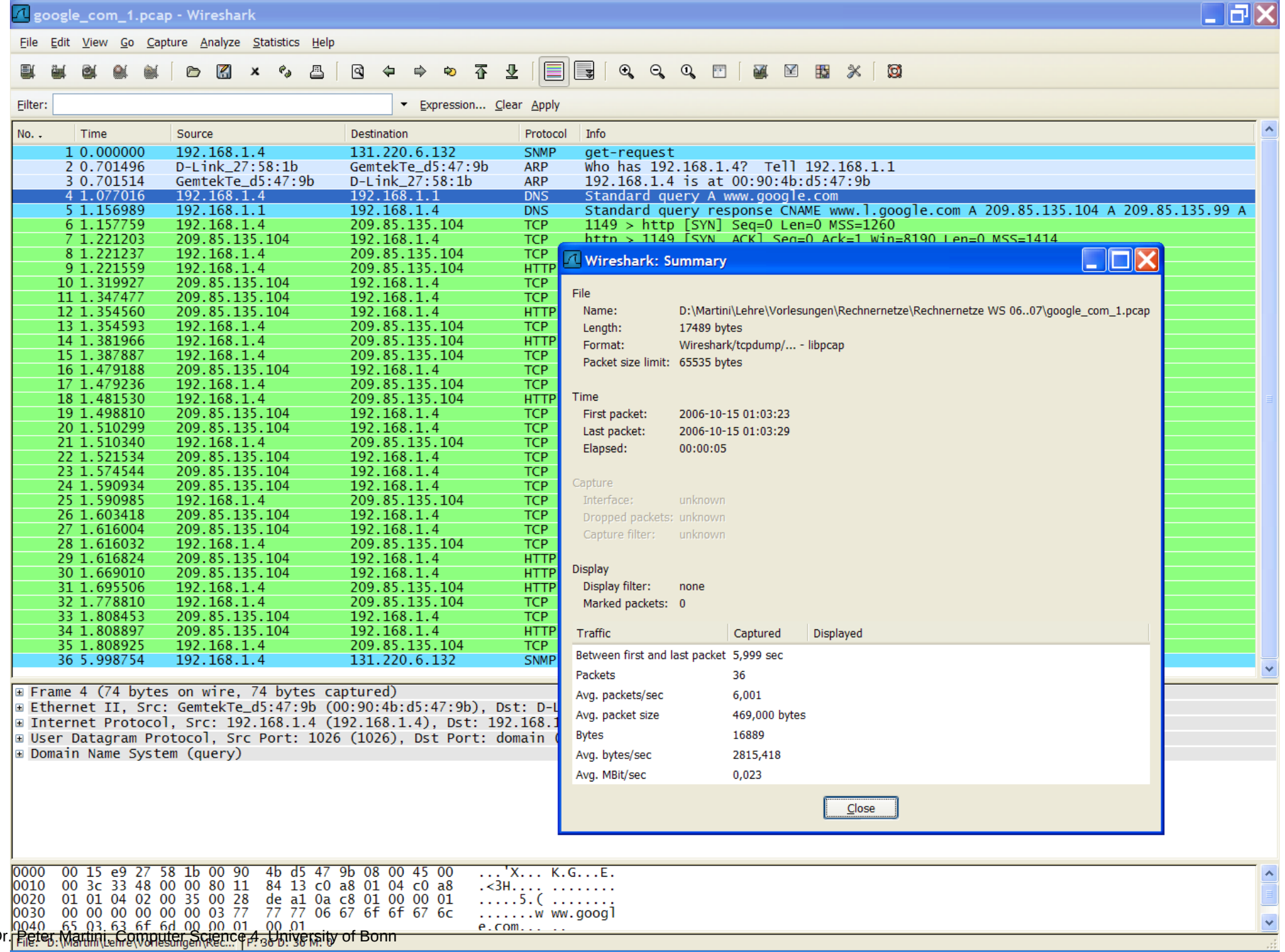
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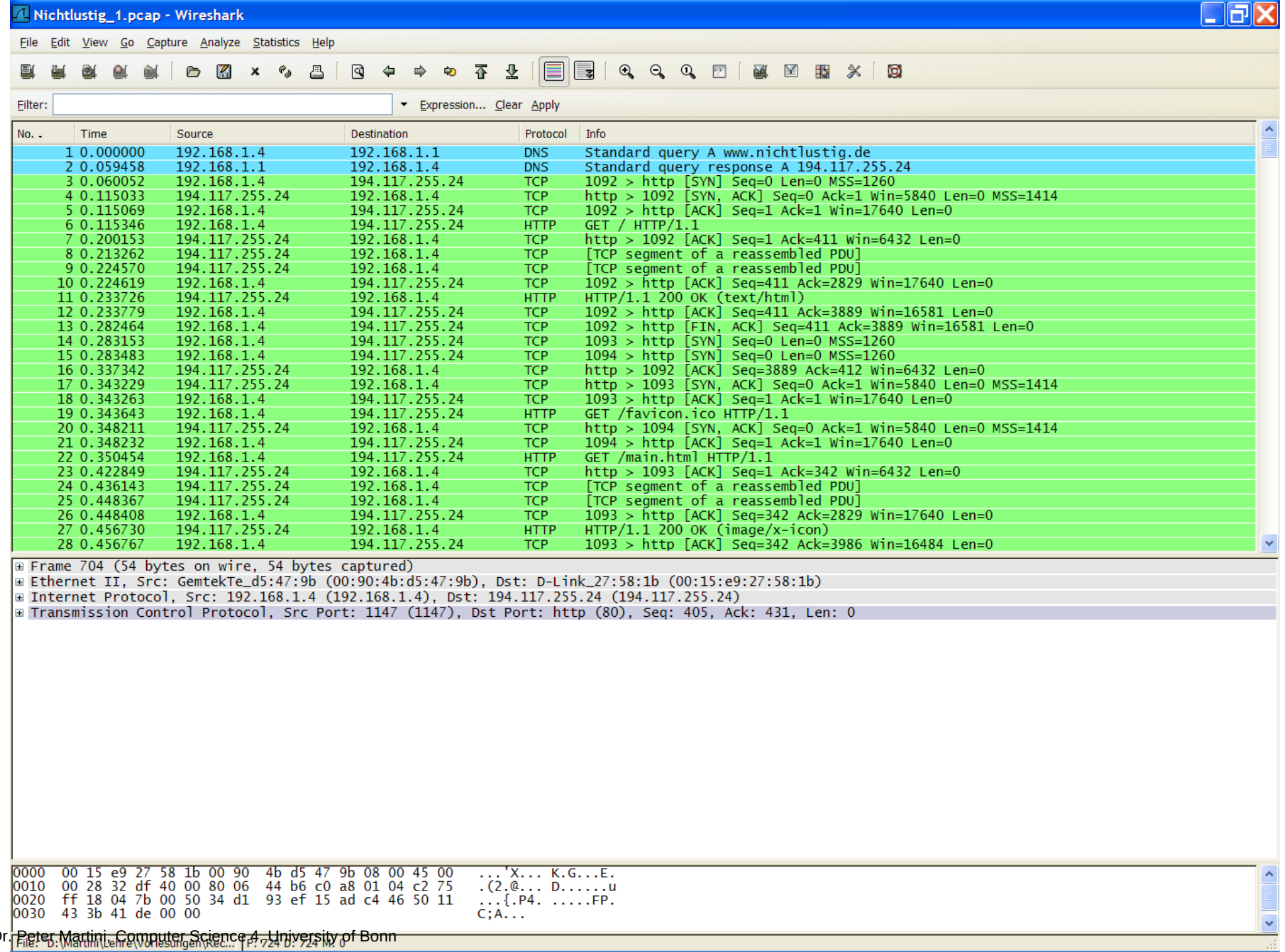
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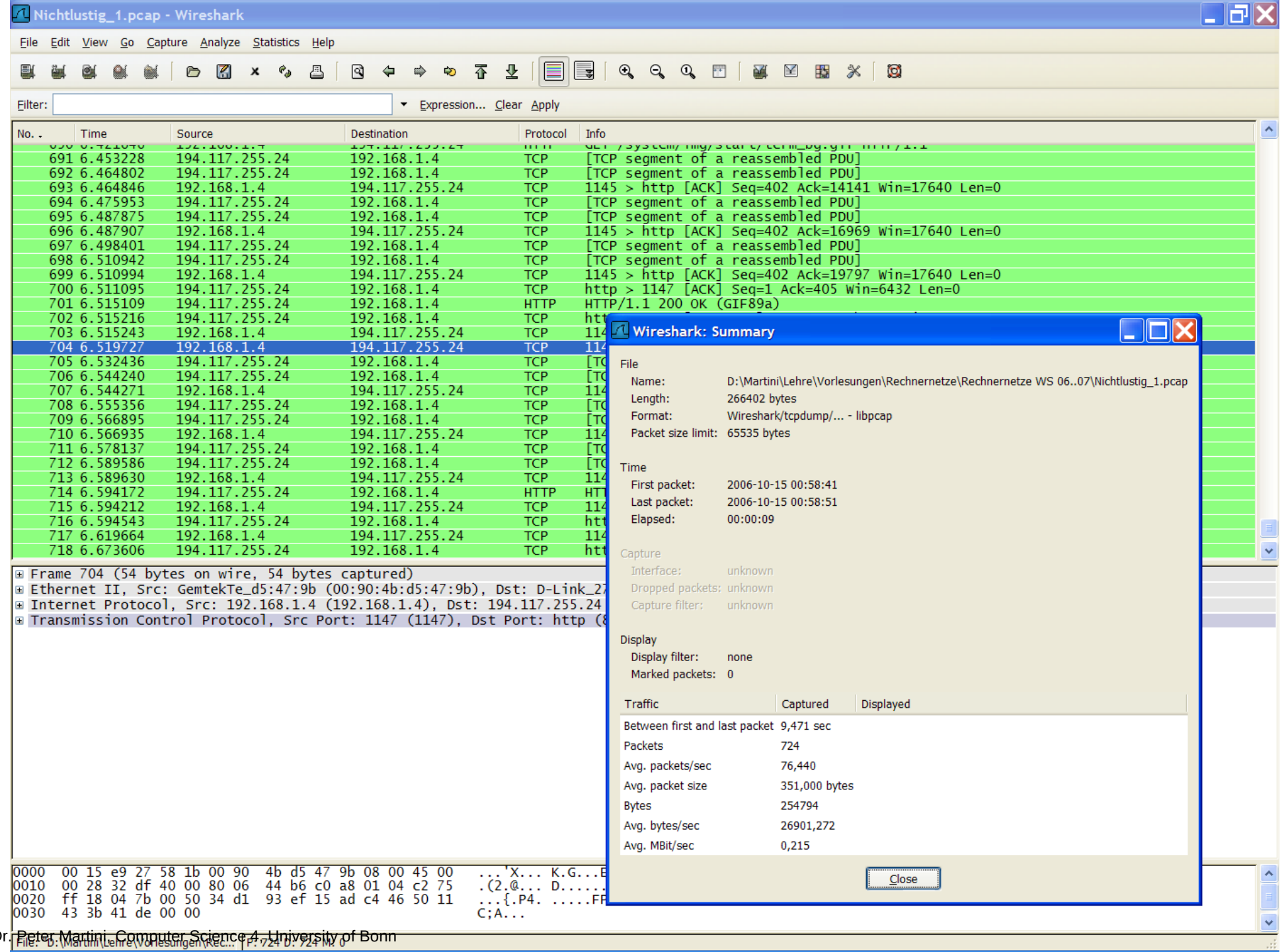
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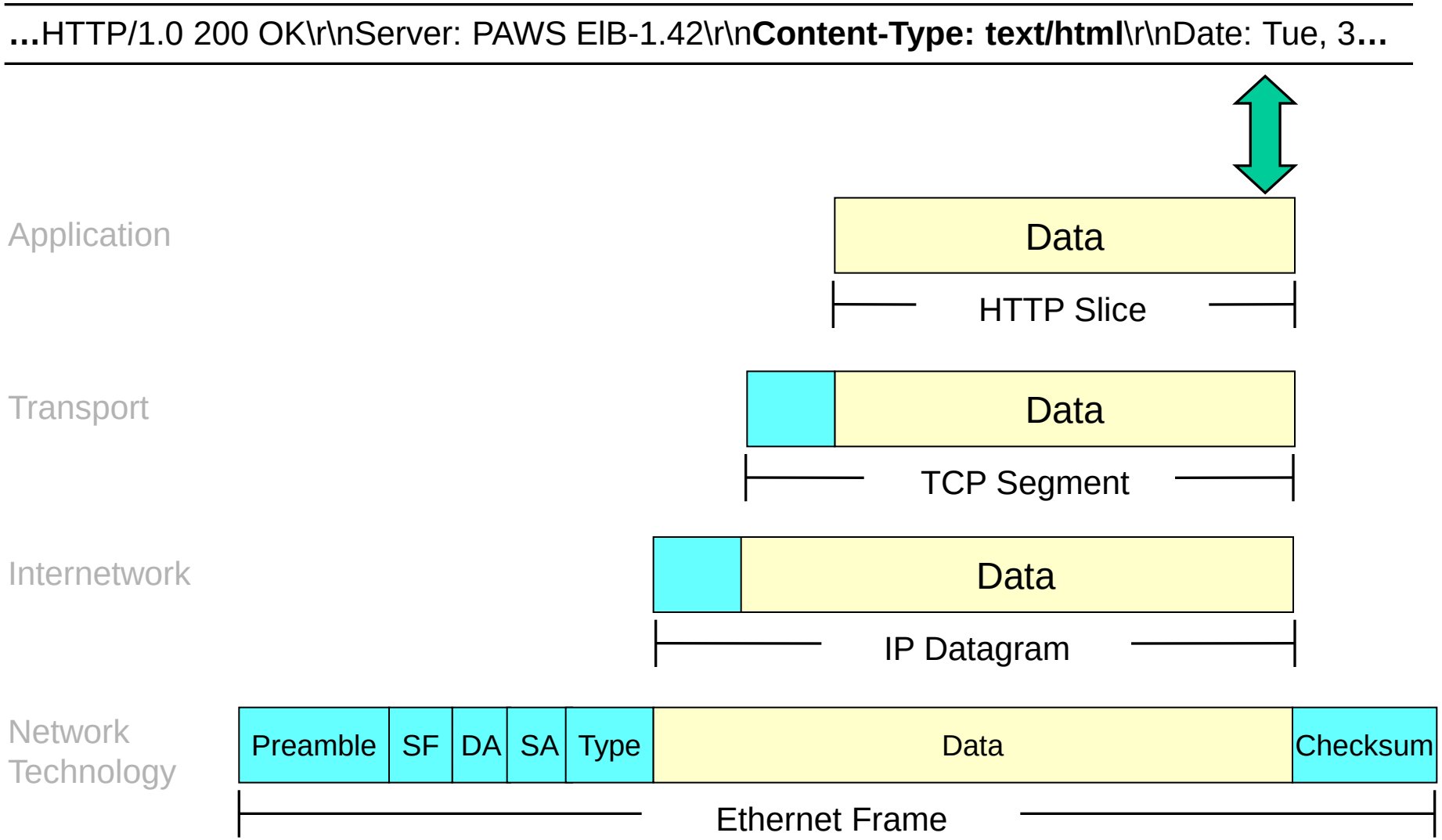


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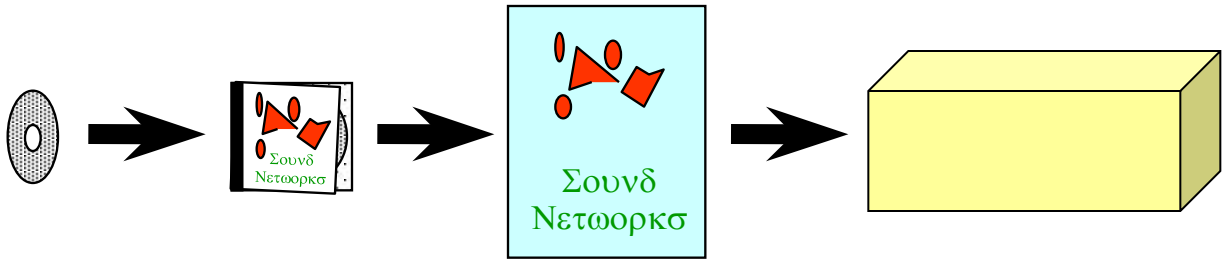
From Ethernet to HTML



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Wrapping of Data

Each layer "wraps" data with an additional envelope (header and/or trailer), before transferring the data to the lower layer:



For the protocol of layer (N) only (N)-PCI is relevant.
The protocol just works on the "wrapping".

But:

- The size of a (N+1)-PDU has to obey certain rules.
  (The postal service does not accept packets of 30 tons)
- The transfer of the PDU has to be done in a proper manner.

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